

Can Inclusionary Zoning Mitigate Climate Gentrification? Income-Based Sorting on Urban Cooling Investments

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Abstract

Cities investing to reduce urban heat face a climate gentrification dilemma. While cooling increases neighborhood amenity values, the rent increases that follow fall hardest on the low-income residents that investments aim to help. Using a non-homothetic spatial equilibrium model of household sorting in Maricopa County, Arizona, I study whether residential zoning reform can moderate this regressive incidence. I find that when implemented alone, Phoenix’s Cool Corridors program provides negligible returns to low-income renters, who keep only \$20 of the \$432 the cooling is worth to them once rents adjust. However, I show that combining cooling with minimum-lot reform that is targeted to the most constrained neighborhoods reverses this effect. Under the combined policy scenario, the low-income gain rises to \$506 a year and the ratio of high- to low-income gains falls from twenty-seven to one, with more low-income residents remaining in improved neighborhoods. Additionally, I show that the use of homothetic demand – conventional in the environmental sorting literature – understates the

affordability effects brought on by neighborhood improvement, with consequences for policy conclusions.

1 Introduction

Extreme heat harms health (Barreca et al., 2016) and economic outcomes (Deryugina & Hsiang, 2014; Graff Zivin & Neidell, 2014). Its burden is uneven within cities: heat varies significantly across neighborhoods with differences in land cover and built form (Hsu et al., 2021), with these gaps widening during heat waves (Li & Bou-Zeid, 2013). Recent work has shown that this variation in urban heat exposure is socioeconomically stratified— in the highest-risk U.S. cities the lowest-income decile of neighborhoods averages 4°F hotter than the highest (Dialesandro et al., 2021). Low-income households are also least able to adapt, bearing greater vulnerability (Mitchell & Chakraborty, 2015) and limiting cooling to afford other necessities (Cong et al., 2022). In response, cities and states have begun investing heavily in neighborhood cooling in their lowest-income areas (US EPA, 2025), among them California’s \$300 million extreme-heat package (2022) (Office of the Governor of California, 2022), New York’s \$106 million Cool Neighborhoods program (City of New York, Mayor’s Office of Recovery and Resiliency, 2017), and Phoenix’s \$60 million Shade Phoenix Plan (City of Phoenix, Office of Heat Response and Mitigation, 2024).

Yet these investments may undermine their own goal. Cooling makes a neighborhood more desirable, and where housing supply is constrained the added demand is absorbed by higher rents rather than new units, pricing out the incumbent low-income residents whom the majority of public investments target. This process has been termed climate gentrification (Thompson et al., 2023). Recent work documents this effect. Borsky et al. (2024) find that households in New York City sort on microclimate and bid up rents in cooler locations following cooling investment, Angelovski et al. (2022) find significant gentrification

from urban greening across 28 cities, and Connolly (2025) warns that such displacement undermines the long-run effectiveness of urban climate action.¹ Public cooling thus poses a distributional dilemma to cities: the same amenity that reduces low-income heat exposure can leave households worse off through the housing market by displacing some and raising the rent burden of those who remain in the improved neighborhood.

In this paper I examine whether residential zoning reform can mitigate the climate gentrification effects of Phoenix’s “Cool Corridors” tree-planting program. I build a general equilibrium model of Maricopa County, which contains the Phoenix metropolitan area. Households of differing income sort across neighborhoods for their amenities and consume housing under non-homothetic preferences and minimum-lot regulations, which, following Kulka (2019) and Macek (2024), act as a floor on the housing a household must consume to enter a neighborhood. Housing prices and supply then adjust to cooling and zoning shocks. The model captures rich decision-relevant heterogeneity across block-group neighborhoods, which differ in housing stock, amenity levels, supply elasticities, and urban heat exposure. To estimate the causal effect of neighborhood temperature on amenity value, I apply an instrument, developed in Kamen (2026), that exploits each neighborhood’s orientation relative to nearby urban hot and cool spots.

I focus on minimum-lot requirements for four reasons. First, where they bind, rising neighborhood demand is absorbed through higher rents rather than new units (Macek, 2024; Song, 2025); thus, zoning reform targets the rent channel through which climate gentrification operates. Second, they are the most common single-family density restriction in the United

¹In addition to empirical work, other evidence suggests that this concern is salient to community members in low income areas. Community action groups have documented persistent displacement and gentrification fears among low-income Phoenix residents in response to public-infrastructure investment, particularly the South Phoenix light-rail extension (Yale Climate Connections, 2023), and Guardaro et al. (2020) likewise record infrastructure-driven gentrification concerns in low-income Phoenix neighborhoods as a barrier to urban greening projects. Low-income residents in other cities facing extreme heat have voiced the same concern about greening directly: in Los Angeles, residents resist tree-planting out of fear it will raise rents (LAist, 2024), similar resistance is documented in Detroit and Philadelphia (Carmichael & McDonough, 2019; Pearsall et al., 2024). This is consistent with quantitative evidence that greening is associated with subsequent gentrification (Anguelovski et al., 2022; Donovan et al., 2021).

States (Song, 2025), and in low-income neighborhoods they likely reflect decisions of past residents.² Third, relaxing them poses little risk of worsening the heat the policy is meant to relieve. As typically enacted, zoning reform permits infill, for example adding units to already-developed single-family lots by converting existing homes into duplexes, triplexes, and fourplexes³(Arizona State Legislature, 2024), rather than extending development onto undeveloped land, rezoning parks or open space, or building large complexes, the land-cover conversions that overwhelmingly drive urban heat (Connors et al., 2013; Imhoff et al., 2010; Stone & Rodgers, 2001). Fourth, reforming binding minimum lot requirements recovers efficiency gains that zoning itself was withholding by allowing households to choose their optimal housing consumption rather than being forced to consume more (Kulka, 2019; Macek, 2024). This mechanism stands in contrast to policies such as low-income housing subsidies or vouchers, which require direct, large transfers (Collinson et al., 2015).

I find that Phoenix’s Cool Corridors program, which costs \$1.4 million annually (City of Phoenix, 2025), leaves low-income renters almost no better off when applied on its own. This is because the rent increases that follow the cooling improvement offset almost the entire value of the cooling itself. However, pairing the same cooling program with a minimum-lot reform targeted to the neighborhoods where existing zoning constraints bind most reverses this outcome, delivering large net welfare gains to low-income renters of \$506 annually and

²The minimum-lot reforms evaluated in this paper are located in low-income neighborhoods, where regulations were often set at development during the middle of the 20th century and the current lower-income occupants, who filtered into the aging housing stock later, did not choose them (Fischel, 2004; Rosenthal, 2014). Opposition to relaxing minimum-lot requirements typically centers on the fear that added density will erode property values, and is overwhelmingly concentrated among homeowners in higher-income communities (Fischel, 2001; Hankinson, 2018). The income composition of this opposition is aligned with empirical results showing that home-value depreciation from deregulation is primarily a concern in wealthy neighborhoods (Macek, 2024); Gyourko and Krimmel (2021) show that the minimum-lot premium is concentrated in high-cost markets and modest but still substantial elsewhere, and Hilber and Robert-Nicoud (2013) that the value of land-use constraints accrues to owners in the most desirable, developed places.

³Infill could in principle add impervious surface where minimum-lot reform is paired with relaxed building setbacks, so that a larger footprint replaces a previously vegetated yard. This concern is negligible in the low-income neighborhoods where cooling investment and this paper’s reforms concentrate, because private vegetation there is typically very sparse (Hope et al., 2003; Leong et al., 2018). In Phoenix, low-income lots tend to be bare or xeric rather than irrigated, with neighborhood tree cover typically 2.5–8% (City of Phoenix, Office of Heat Response and Mitigation, 2024).

drawing more of them into the cooled neighborhoods. Further, I show that these gains do not come at the expense of incumbent homeowners. While relaxing the minimum lot on its own lowers home values, pairing it with the cooling improvement leaves owner-occupiers better off, assuaging the property value loss concern that high income opponents typically express. Lastly, I find that the conventional homothetic demand system attributes regressivity of cooling amenity improvements almost entirely to differences in preferences by income group— which housing policy cannot change, rather than to the affordability channel through which climate gentrification operates— which housing policy can offset. This result is meaningful for policy decisions because a planner relying on it would observe a weaker case for implementing a complementary housing policy than if they relied on analyses using non-homothetic demand.

This paper makes three main contributions. First, it is, to my knowledge, the first to bring residential zoning and urban climate adaptation into a single equilibrium framework, and my results show that climate gentrification is not an unavoidable consequence of cooling but instead depends significantly on the housing-supply environment in which cooling is delivered. This work relates to both urban policy work and the environmental sorting literature. The urban economics literature establishes that restrictive residential zoning raises housing costs and constrains supply (Glaeser & Gyourko, 2003; Gyourko & Molloy, 2015; Hsieh & Moretti, 2019; Turner et al., 2014), and that relaxing these constraints lowers rents and delivers progressive gains to low-income renters (Büchler & Lutz, 2024; Macek, 2024). Meanwhile, the environmental sorting literature studies how households sort in response to environmental risk, typically omitting the impact of housing policy design (Bakkensen & Ma, 2020; Borsky et al., 2024; Fan et al., 2018; Wrenn, 2024). Only Ospital (2022) models environmental sorting jointly with housing-supply policy, and they focus on wildfire risk at the urban fringe. No prior work examines how zoning mediates exposure to urban heat, or identifies zoning reform as a tool that cities facing affordability crises can use to contain climate gentrification. I address this gap by showing that cooling investment and zoning

reform are complementary policy tools. Enacting them together lets the welfare gains from cooling reach low-income households rather than being capitalized into rents.

Second, I show that researchers’ choice of demand system in sorting models matters considerably for distributional policy analysis. Sorting models in the environmental literature commonly assume homothetic, Cobb-Douglas preferences over housing (Diamond, 2016; Fan et al., 2018; Wrenn, 2024). In this demand specification, every household spends a constant share of income on housing, so an amenity-driven rent increase burdens rich and poor equally as a fraction of income. Non-homothetic Stone-Geary preferences instead reproduce the observed fall in the housing-expenditure share as income rises, from about 0.42 at the lowest income to 0.13 at the highest in the Phoenix metro area, so the same rent increase falls disproportionately on the poor. Because the housing expenditure income gradient is built into the demand system instead of fitted to location choices, a non-homothetic model can separate the welfare effect of neighborhood amenities from that of rent changes. I find that this distinction is quantitatively large. Overall, I show that homothetic demand flattens the welfare incidence of cooling alone across the income spectrum. It understates the annual price effect from rent increases on the lowest-income renters (\$289 versus \$412 annually)—while overstating the value those households place on relaxing the minimum lot by 160 percent (\$1,156 against \$444). Thus a planner relying on this demand system would both underestimate the extent to which affordability pressures fall on low income residents and overstate the extent to which zoning reform helps them.

Third, the paper shows that the joint spatial incidence of cooling and zoning reform determines their distributional outcome, helping to inform how such policies— which are often set separately from each other (Keith et al., 2025) — should be designed. An example of this mismatch is evident in Arizona’s recent HB 2721 policy, which relaxes minimum lots within a mile of city centers (Arizona State Legislature, 2024), an area that overlaps negligibly with the hot, low-income neighborhoods that Phoenix’s cooling programs target, and which are

susceptible to rent increases. Furthermore, I find that the gains from reform are concentrated in the neighborhoods where the minimum lot binds hardest, a fraction of the area the cooling policy covers. Thus, the spatial extent over which zoning reform should be enacted in order to protect low income renters from price increases depends on the distribution of low income households across neighborhoods in which both lot requirements bind and amenity improvement policies are applied.

This paper’s central discussion focuses on policy impacts on low income renters. While I estimate welfare impacts to owner-occupiers, this group faces different tradeoffs than renters. Owner-occupiers gain unambiguously from the cooling policy alone, which raises home values through the amenity improvement with no offsetting supply response, and they gain under the combined policy as well. This is because zoning reform on its own does lower home values where zoning binds, but only slightly, because the reform’s large gains to renters come from the quantity margin instead of the price margin. The reform lets constrained households stop over-consuming housing and re-optimize their bundle, an efficiency gain that is created instead of transferred out of owners’ home values⁴. As a result, the small annual capital loss from zoning reform is more than offset by the capital gain from cooling. On the other hand, low-income renters are the households most exposed to the rent increases that follow gentrification, through either a higher housing burden or displacement. Renters further make up the majority (63.7%) of low income households in the neighborhoods that Phoenix’s Cool Corridors program targets.

Although this work focuses on the Phoenix metro area, results are applicable to many other U.S. cities. While Phoenix has historically faced some of the hottest temperatures,

⁴The home-value effect of relaxing a minimum lot is small in low-income neighborhoods because the constraint capitalizes little into prices there to begin with (this is compatible with large efficiency gains from low-income households being able to re-optimize into smaller housing bundles because that gain comes from households recovering what they were forced to overspend on unwanted housing, while the space smaller units free up is taken up by new residents moving in rather than left vacant to push prices down). The minimum-lot premium is therefore concentrated in high-cost markets (Gyourko & Krimmel, 2021) and the value of land-use constraints accrues to owners in the most desirable, developed places (Hilber & Robert-Nicoud, 2013).

severe heat exposure is not unique to the metro area. Heat waves have grown more frequent, longer, and more intense across large U.S. cities over the past half-century (Habeeb et al., 2015), and extreme humid heat, measured by wet-bulb temperature, has more than doubled in frequency since 1979, with especially severe incidence across the southeastern United States and parts of the Midwest (Raymond et al., 2020). Additionally, extreme heat day changes brought on by climate change over the next several decades are projected to extend such conditions across more regions (Dahl et al., 2019). Further, heat inequity is ubiquitous across the U.S., with significant disparities recorded in 175 major U.S. cities (Hsu et al., 2021). The zoning policy results from this paper also extend across urban centers in the U.S., with minimum lot requirements being the most common form of density regulation in the country (Song, 2025), and binding supply constraints present in most large U.S. housing markets (Glaeser & Gyourko, 2003; Turner et al., 2014). These conditions are reflected by the current policy environment, in which many cities have recently passed, or residents have begun to advocate for, similar forms of zoning de-regulation (Manville et al., 2020).

Further, the framework I develop applies beyond urban heat relief programs. Amenity improvements capitalize into rents wherever housing supply is inelastic (Been et al., 2019; Gyourko & Molloy, 2015), and the resulting burden falls disproportionately on the poor because housing is a necessity. These conditions define many contexts in which improvements made in low income neighborhoods spur tension between the residents they are meant to help and the corresponding housing cost increases (Diamond, 2016). This paper shows that whether or not these investments make low income residents meaningfully better off depends on the housing-supply policies that govern its incidence, and a complementary supply reform targeted to where supply is constrained can help to ensure that the benefits of neighborhood improvements flow to the intended low income recipients. Climate-adaptation and housing-supply policies are typically set through separate processes, so this research emphasizes the importance of policymakers recognizing and implementing them as complementary tools.

2 Empirical Overview

2.1 Urban Heat and Cooling Policy in Maricopa County

I study household sorting in Maricopa County, Arizona, which contains the Phoenix–Mesa–Chandler metropolitan core and is home to roughly 1.7 million households.⁵ Extreme heat is the county’s foremost environmental concern: average July temperatures have risen past 100°F in recent years (US Department of Commerce, 2025), and residents perceive intra-city differences at fine spatial scale, registering microclimate variation across their own neighborhoods (Jenerette et al., 2016).

Crucially for this paper, heat exposure is unevenly distributed and inversely related to income. Figure 1 maps median household income against mean July nighttime minimum air temperature, the metric of urban heat island intensity used throughout the paper; the central urban core and rapidly developed southeastern districts combine the lowest incomes with the highest temperatures, while cooler, well-vegetated areas are systematically higher-income. This within-county income–heat gradient, which motivates the model, mirrors national evidence that the lowest-income neighborhoods are substantially hotter than the highest (Benz & Burney, 2021; Dialesandro et al., 2021; Hsu et al., 2021).

Phoenix has responded with public cooling investment, establishing an Office of Heat Response and Mitigation in 2021, launching the Cool Corridors Program in 2022, and approving a \$60 million Tree and Shade Master Plan in 2024 (City of Phoenix, 2025). I use Cool Corridors as the concrete intervention whose distributional incidence I evaluate.⁶ The city selects corridors through a multi-criteria ranking that prioritizes heat vulnerability, transit depen-

⁵The county’s residential stock has a combined property value exceeding \$1 trillion (Maricopa County Assessor’s Office, 2026).

⁶Administered by the Street Transportation Department, the program has received \$1.4 million in annual Capital Improvement funding since 2022, allocated primarily to planting drought-tolerant trees (roughly 200 per mile) along quarter- to half-mile segments of heavily traveled pedestrian and cyclist routes (City of Phoenix, 2025).

dence, and pedestrian activity (City of Phoenix, 2025), directing cooling disproportionately toward the hottest, lowest-income neighborhoods identified in Figure 1. This targeting makes Cool Corridors a natural setting for studying whether a cooling program, absent complementary housing policy, imposes rent burdens on the residents it is meant to help.

2.2 Residential Zoning Policy in Maricopa County

The housing stock the model describes is predominantly single-family and subject to minimum-lot regulation. Single-family homes make up 67 percent of Phoenix’s housing supply (City of Phoenix, 2024b), and 80.4 percent of arms-length residential sales in Maricopa County between 2014 and 2019 were single-family structures subject to a minimum-lot requirement (Maricopa County Assessor’s Office, 2021). This prevalence is not particular to Phoenix. Single-family requirements cover the large majority of residentially zoned land in most large U.S. cities (Manville et al., 2020), and Song (2025) finds that 18.5 percent of single-family construction nationwide bunches at the minimum-lot threshold, so these requirements bind for a substantial share of development. Phoenix is representative of the regulatory environment facing most U.S. metropolitan housing markets.

Arizona has already begun enacting small reform measures. Arizona’s “Middle Housing” bill (HB 2721; A.R.S. § 9-462.13), passed in 2024 and effective January 2026, requires municipalities of more than 75,000 residents to permit duplexes, triplexes, fourplexes, and townhomes by right on single-family lots within one mile of their central business districts, and to open at least 20 percent of lots in large new subdivisions to these types (Arizona State Legislature, 2024). By relaxing the single-family use restriction, such reforms raise the elasticity of housing unit supply, so that added demand is met more through new units and less through higher rents. I do not evaluate the bill’s own distributional effect. The spatial incidence of HB 2721 has negligible spatial overlap with the Cool Corridors, so I

fold Phoenix’s proposed compliance into the baseline against which the counterfactuals are measured. However, the passage of HB 2721 establishes is that type of infill development or, ”middle-housing reform” is a viable policy instrument in this setting. in this paper I examine where such reform should be aimed, and how it interacts with public cooling investment. Because the reform postdates the data, the minimum-lot measures I use, mapped in Figure 2, reflect the regulatory environment as of December 2025 and serve as the pre-reform baseline.

Minimum lots are smaller in the urban core and in the neighborhoods the program targets, though still substantial and overwhelmingly limited to single family dwellings. The mean minimum lot is 7,277 square feet countywide, 6,351 within Phoenix, and 6,202 across corridor block groups, close to Phoenix’s smallest single-family district (R1-6, 6,000 square feet). Historically The most historically high-income neighborhoods have the most stringent single-family zoning (Kulka, 2019), but minimum-lot regulation remains pervasive across lower-income areas as well, for two reasons. Zoning is largely fixed at development and revised infrequently, so a designation can persist for decades (Song, 2025). The single-family stock built under these rules filters down to lower-income occupants as it ages (Rosenthal, 2014), so neighborhoods originally developed for middle-income single-family housing come to house low-income residents while retaining their minimum lots. Many low-income, heat-vulnerable households in Phoenix are therefore concentrated in built-out neighborhoods where land is developed and densification is restricted.

That Cool Corridor neighborhoods overlap substantially with minimum-lot dominated neighborhoods, only 22 percent of corridor mileage lies in multifamily-zoned block groups free of a minimum lot. Permitting additional units on these lots, as HB 2721 does, is the margin on which supply can expand to absorb cooling-induced demand rather than pass it through to rents. Further motivating the use of zoning reform in these neighborhoods, supply in these neighborhoods is inelastic. Figure 3 overlays the corridors on block-group supply elasticities from Baum-Snow and Han (2024): the median unit supply elasticity is

0.42 countywide, 0.37 within Phoenix, and 0.35 across the low-income corridor neighborhoods, among the most constrained in the metropolitan area. Where supply cannot expand, an amenity improvement raises the housing-cost burden on incumbents rather than drawing new construction, a condition under which a supply-side reform can keep the improvement from being capitalized into rents.

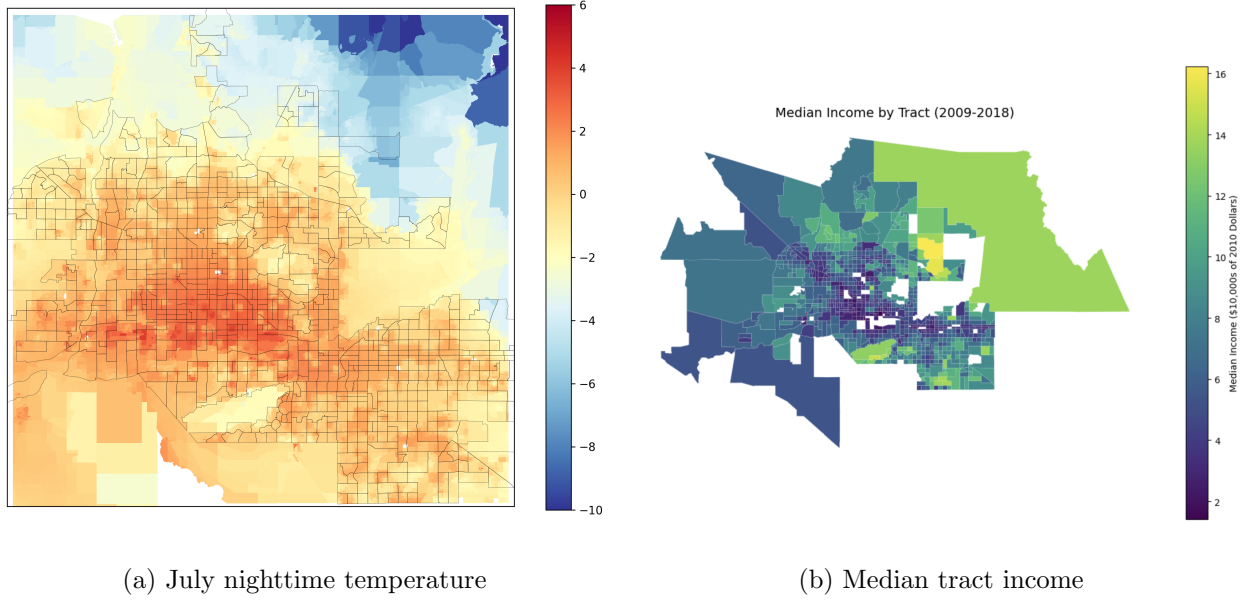


Figure 1: **Income and urban heat in Maricopa County.** Panel (a) maps July nighttime minimum air temperature by census block, expressed as the deviation ($^{\circ}\text{C}$) from the countywide mean; panel (b) maps median tract income (in \$10,000s). The hottest areas, in the central urban core and the rapidly developed southeast, coincide with the lowest incomes, while cooler, well-vegetated areas are systematically higher-income. Temperature data are from NCAR’s High-resolution Urban Meteorology for Impacts Dataset (HUMID).

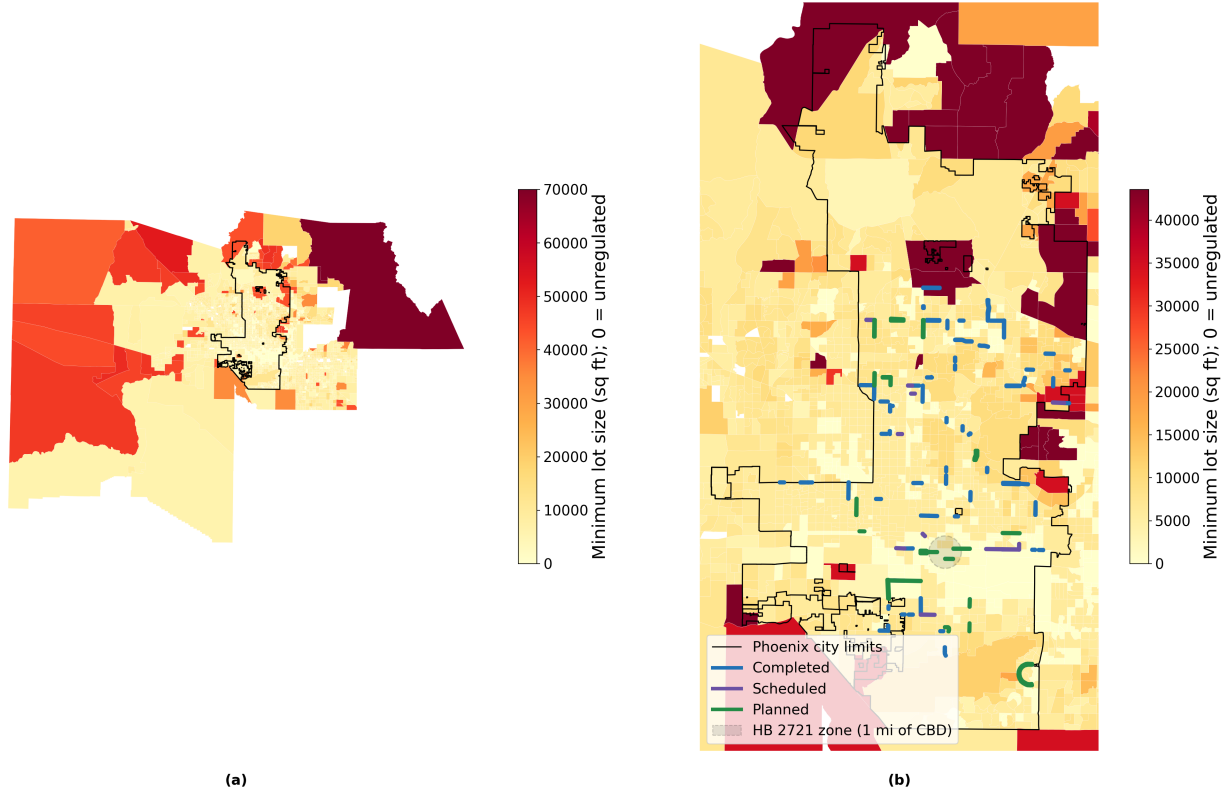


Figure 2: Minimum lot-size requirement by block group, as of December 2025. Panel (a) shows Maricopa County; panel (b) restricts the view to Phoenix and overlays the Cool Corridors, distinguished by project status (completed, scheduled, and planned). Values of zero denote block groups zoned for multifamily housing with no single-family minimum-lot requirement. In panel (b), the shaded circle marks the one-mile radius around Phoenix's downtown core within which the currently enacted HB 2721 middle-housing reform permits duplex- through fourplex-development by right; its limited overlap with the Cool Corridors illustrates why relaxing zoning under the enacted reform alone does little to relieve rent pressure in the cooled neighborhoods, motivating the targeted reform evaluated in the counterfactual analysis in this paper. Very high minimum lots are concentrated in the county's northeastern and western peripheries. The former comprises the desert foothills of Scottsdale, Cave Creek, and Carefree, where large-lot and environmentally-sensitive-lands zoning preserves low-density desert character in some of the metro-areas highest-income communities. The latter consists of unincorporated, or agricultural desert, reflecting exurban land use rather than the urban housing market.

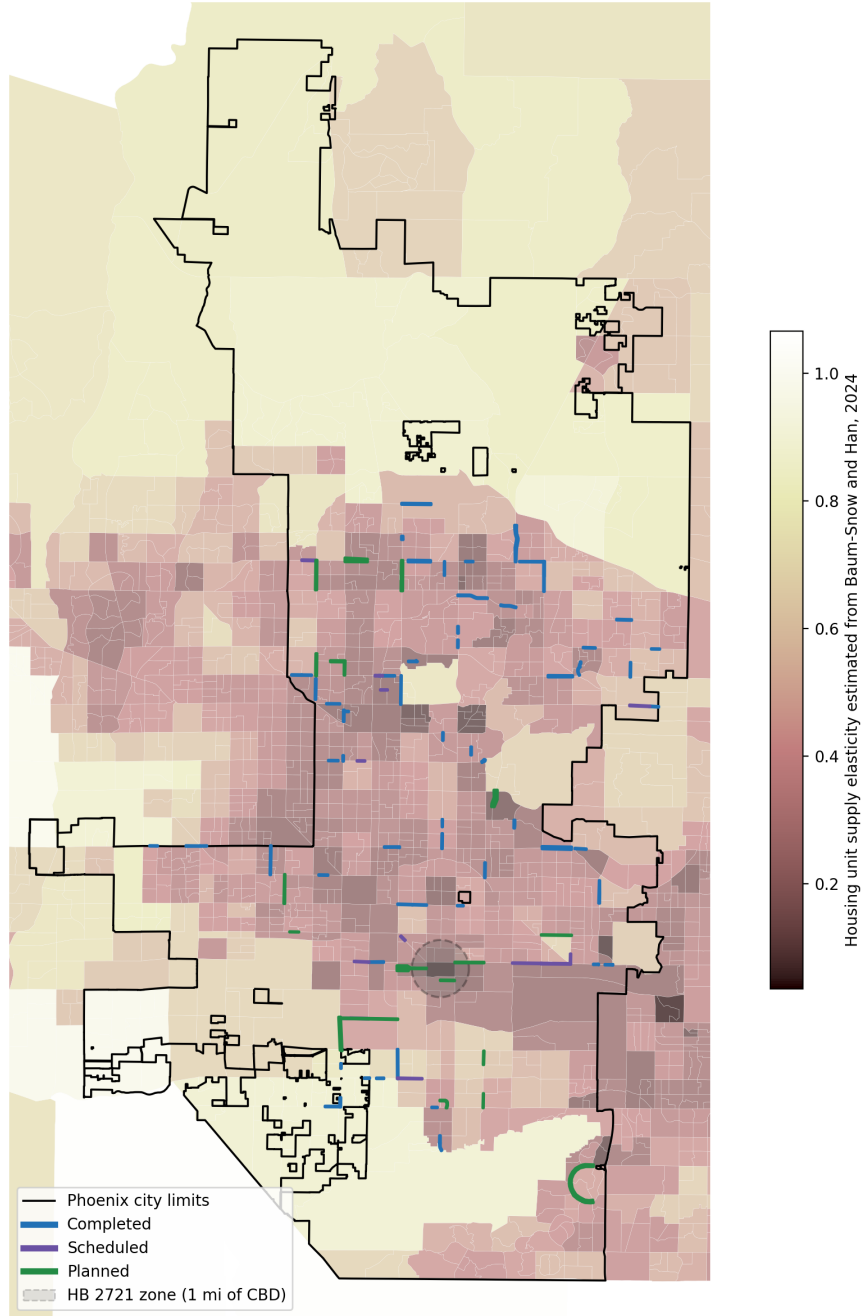


Figure 3: Estimated housing unit supply elasticity by block group within Phoenix, from Baum-Snow and Han (2024), with completed, scheduled, and planned Cool Corridors overlaid. Darker shading indicates lower elasticity, (more supply-constrained) neighborhoods. The dashed circle marks the one-mile radius around the downtown core within which the current HB 2721 permits middle housing. Elasticities are estimated at the tract level and assigned to constituent block groups, so variation is constant within tracts.

3 A Two-Sided Spatial Sorting Model

I develop a two-sided general equilibrium model of residential sorting to determine how zoning regulation shapes household responses to an improved urban microclimate. In the model, households trade off neighborhood amenities for housing costs when choosing where to live, subject to a minimum lot requirement, while landowners supply housing in response to neighborhood supply elasticities and household demand. The framework adapts the sorting model of Macek (2024), who evaluates the welfare effects of national housing supply deregulation in the United States. Applied in a local setting, I extend this model to measure the distributional welfare effects of residential minimum-lot constraints and neighborhood improvements in low-income microclimates across the Phoenix metro area.

3.1 Setup

Maricopa County consists of N neighborhoods, delineated by census block groups and indexed by n . It is populated by heterogeneous renting households that fall into W income groups, indexed by w . Each household chooses a neighborhood n in which to reside and, within that neighborhood, a zone $o \in \{\text{reg}, \text{unreg}\}$. The regulated zone is subject to a minimum lot requirement; the unregulated zone is zoned for multifamily housing and carries no minimum lot. Households consume the local housing good h in zone o of neighborhood n , a freely traded homogeneous numeraire good g , and enjoy an amenity level a_{wn} . On any parcel in zone o of neighborhood n , absentee landowners maximize profits with a Cobb-Douglas production technology over a fixed land input and a capital input obtained from the numeraire good at a rate of transformation of unity.

3.2 Household Utility

Households have Stone-Geary preferences over the local housing good h and the numeraire good g . A household of group w living in zone o of neighborhood n solves

$$V_{o,wn} = \max_{h,g} \underbrace{k_w \beta^{-\beta} (1-\beta)^{-(1-\beta)} (h - \bar{h})^\beta g^{1-\beta}}_{\text{Stone-Geary consumption value}} + \underbrace{\log a_{wn}}_{\text{amenity}}, \quad (1)$$

subject to

$$P_n^o h + g \leq I_w, \quad P_n^o h \geq R_n \text{ if } o = \text{reg}, \quad h \geq \bar{h}, \quad g \geq 0. \quad (2)$$

where $\bar{h}$ is a subsistence level of housing common to all households, I_w is the mean income of group w , R_n is the cost of the minimum lot in the regulated zone, and k_w is an income-group-specific weight on the consumption value that adjusts for differences in consumption scale across income groups. In the unregulated zone, or a regulated zone where the minimum lot does not bind, household indirect utility takes the form

$$V_{o,wn} = \log a_{wn} + k_w \frac{I_w - P_n^o \bar{h}}{[P_n^o]^\beta}, \quad (3)$$

and housing consumption at the optimum is

$$h_{wn}^* = \bar{h} + \frac{\beta}{P_n^o} (I_w - P_n^o \bar{h}). \quad (4)$$

When the minimum lot binds, housing consumption is fixed at R_n/P_n^{reg} and numeraire consumption at $I_w - R_n$. After substituting these terms into the household utility function, the household's utility can be written as its undistorted consumption value scaled by a

distortion factor D_{wn} , plus the additive amenity (Macek, 2024):

$$V_{\text{reg},wn}^{\text{bind}} = \log a_{wn} + \underbrace{k_w \frac{I_w - P_n^{\text{reg}} \bar{h}}{[P_n^{\text{reg}}]^\beta}}_{\text{undistorted consumption value}} \times \underbrace{\kappa \frac{(R_n - P_n^{\text{reg}} \bar{h})^\beta (I_w - R_n)^{1-\beta}}{I_w - P_n^{\text{reg}} \bar{h}}}_{D_{wn}}, \quad \kappa \equiv \beta^{-\beta} (1-\beta)^{-(1-\beta)}. \quad (5)$$

Under the conditions implied by the model's structure, $0 < \beta < 1$, $I_w - P_n^{\text{reg}} \bar{h} > 0$, and $P_n^{\text{reg}} \bar{h} < R_n < I_w$, the factor $D_{wn} \leq 1$ is a penalty to utility from the binding lot. Taking the derivative of D_{wn} with respect to income reveals that the penalty is largest for low income households:

$$\frac{\partial D_{wn}}{\partial I_w} = \kappa \frac{(R_n - P_n^{\text{reg}} \bar{h})^\beta}{(I_w - P_n^{\text{reg}} \bar{h})^2} (I_w - R_n)^{-\beta} \left[(1-\beta)(R_n - P_n^{\text{reg}} \bar{h}) - \beta(I_w - R_n) \right]. \quad (6)$$

The factor outside the bracket is unambiguously positive, and the bracket is positive as income approaches the cost of the minimum lot R_n . Thus D_{wn} rises with income near R_n , lower income households are more likely to be constrained by the minimum lot requirement and bear the largest distortion when it binds.

3.3 Utility Response to Neighborhood Cooling

In this section I extend the household model to characterize how a marginal improvement in a neighborhood's microclimate, expressed as an increase in amenity value, changes household welfare in equilibrium. Let $\theta_w \geq 0$ denote the direct utility benefit to a group- w household of a one-degree reduction in temperature, which operates through the amenity term a_{wn} . Because cooling makes the neighborhood more attractive, its equilibrium housing price rises, imposing a rent penalty $\Phi_{wn} \geq 0$ that offsets the direct cooling benefit.

In an unregulated zone, or a regulated zone where the minimum lot does not bind, the equilibrium welfare change from marginal cooling improvement is the direct cooling benefit

net of the rent penalty,

$$\Delta V_{\text{unreg}, wn}^{\text{cool}} = \underbrace{\theta_w}_{\text{direct benefit}} - \underbrace{\Phi_{wn}}_{\text{rent penalty}}, \quad \Phi_{wn} = -\frac{\partial V_{wn}^{\text{unreg}}}{\partial P_n^{\text{unreg}}} \frac{\partial P_n^{\text{unreg}}}{\partial(\text{cooling})}. \quad (7)$$

Under non-homothetic demand, the rent penalty is proportional to the household’s housing expenditure share (Appendix A), and is larger for low-income households who spend greater share of income on housing. Thus, the likelihood that the harm from the rent increase eats away significantly from the cooling benefit or even surpasses it in magnitude is higher for low-income households, even though they value the cooler climate. This effect is the driving force behind the regressive incidence of cooling reported in Section 6.2.1. In my results, the direct benefit appears there as the cooling-amenity component of equivalent variation, and the rent penalty as the net price effect.

In neighborhoods where the minimum lot binds, cooling raises the per-unit price and therefore the cost of the mandated lot R_n , and changes the distortion the lot requirement imposes. Appendix A reports this full composition and shows that this channel is most costly for lower-income households. A binding minimum lot therefore amplifies the regressive incidence of cooling for the low-income households I target in the counterfactuals, which motivates my counterfactual in which I pair the two policies, Cool Corridors and zoning reform, in Section 6.2.1.

3.4 Neighborhood Choice

Each neighborhood n offers a group-specific amenity value a_{wn} to households of group w who reside there, and these amenity values rationalize the observed distribution of population and income across neighborhoods in Maricopa County. This structure allows me to simulate a shock to neighborhood microclimates, delivered by public investment in urban heat mitigation, as a change in a_{wn} .

A household first chooses a neighborhood n and then, conditional on n , a zone $o \in \{\text{reg}, \text{unreg}\}$ within it, where the regulated zone is subject to the minimum lot requirement and the unregulated zone is not. Let $V_{o,wn}$ denote the deterministic utility of a group- w household in zone o of neighborhood n , defined in Section 3.2. Households draw i.i.d. Type-I Extreme Value (Gumbel) shocks at both levels, yielding a nested logit structure. the within-neighborhood inclusive value over the two zones is defined as

$$\Omega_{wn} \equiv \left(\sum_{o \in \{\text{reg}, \text{unreg}\}} \exp(V_{o,wn})^{\rho_{\text{within}}} \right)^{1/\rho_{\text{within}}}, \quad (8)$$

, which summarizes the value of the housing options in neighborhood n to a group- w household. Conditional on residing in n , the household chooses zone o with probability

$$s_{o|wn} = \frac{\exp(V_{o,wn})^{\rho_{\text{within}}}}{\sum_{o' \in \{\text{reg}, \text{unreg}\}} \exp(V_{o',wn})^{\rho_{\text{within}}}}, \quad (9)$$

and it chooses neighborhood n with probability

$$s_{wn} = \frac{(\Omega_{wn})^\rho}{\sum_{n'} (\Omega_{wn'})^\rho}. \quad (10)$$

The unconditional probability that a group- w household resides in zone o of neighborhood n is $s_{o,wn} = s_{wn} s_{o|wn}$. The parameter $\rho > 0$ disciplines the migration response between neighborhoods and $\rho_{\text{within}} > 0$ the response between zones within a neighborhood, with $\rho < \rho_{\text{within}}$, reflecting that housing options between zones within a neighborhood are closer substitutes than options across neighborhoods over the metro area.⁷

⁷I set $\rho_{\text{within}} = 8.5$, the elasticity of substitution between housing options within the same neighborhood, from Macek (2024), and $\rho = 3.9$, the elasticity of substitution between neighborhoods across the multiple municipalities of U.S. metropolitan areas, from Baum-Snow and Han (2024).

3.5 Welfare for Owner-Occupiers

Every household in the model solves the renter’s problem and sorts across neighborhoods accordingly. I then impose the effects of tenure in partial equilibrium, after the equilibrium is solved ⁸, using ownership rates ϕ_{wn} from the ACS sample data. An owner-occupier then receives the change in the value of the home it held at the baseline. For a fixed quantity of owned housing h_{wn}^{own} , this is expressed as the change in the home’s annual rental value,

$$G_{wn} = h_{wn}^{\text{own}} (P_n^{\text{cf}} - P_n^{\text{base}}), \quad (11)$$

which capitalizes into a change in the sale price of G_{wn}/r at a rate r . Because an owner cannot reduce the size of a home it already occupies, G_{wn} is a price effect applied to a fixed quantity, and it is distinct from the renter’s net price effect, which also reflects the quantity the renter re-chooses when prices move.

How an owner’s welfare responds to a policy change depends on what it does with G_{wn} and on whether it also enjoys the new cooling amenity and re-optimizes its housing. While these decisions by owner-occupiers are not endogenous to the model, I report four cases that span these possibilities, shown in Table 5. An owner who *stays and does not refinance their home* receives the amenity gain alone, evaluated at baseline prices and lots. An owner who *sells and leaves the metropolitan area* receives G_{wn} alone. An owner who *borrowes against the home and stays* receives both the amenity gain and G_{wn} . Lastly, an owner who *sells and re-optimizes its housing* is credited with the full renter equivalent variation, including the value of removing the zoning distortion and the net price effect, plus G_{wn} . The capital gain

⁸I do not fully endogenize these gains due to computational limitations. Owners’ capital gains depend on the value of their home in their benchmark location, while sorting decisions also depend on the destination neighborhood. Endogenizing owner gains therefore requires indexing each owner by both origin and destination. The equilibrium then becomes a fixed point over the full origin-by destination distribution of owners rather than over neighborhood populations alone, squaring the dimension of the sorting problem in the number of neighborhoods. I show that the sorting effects this omits are small later in this section.

G_{wn} is credited on the home the household owned at the baseline ⁹.

3.6 Homothetic Demand and Distributional Bias

In this subsection I show that conventional homothetic Cobb-Douglas housing demand distorts each of the three components into which a household’s welfare change decomposes in the policy context studied in this paper. That is, the value households places on the cooling amenity, the burden they bear from the price changes that follow, and the gains that low income groups receive from relaxing the minimum lot requirement. These distortions result from the inability of Cobb-Douglas demand to express housing as a necessity– it does not replicate the reality that low income households spend a larger share of their income on housing than high income households. Under non-homothetic Stone-Geary preferences, the housing expenditure share is $b_{wn} = \beta + (1 - \beta)P_n^o \bar{h} / I_w$, which falls with income. Cobb-Douglas demand sets $\bar{h} = 0$, so the income-varying term $(1 - \beta)P_n^o \bar{h} / I_w$ dissolves and the implied expenditure share is constant at β . I show that this difference in demand structure causes Cobb-Douglas demand to understate both housing price burdens and amenity preferences for low-income households, while overstating these welfare effects for high-income households. An additional implication of this shortcoming, is that homothetic Cobb-Douglas preferences overstate the benefits of zoning reform relative to the non-homothetic Stone-Geary specification.

⁹This approach to expressing owner welfare cannot account for sorting effects driven by the receipt of capital gains themselves. However, because capital gain is represented as a scalar addition to income that raises an owner’s demand for every neighborhood equally rather than making any single neighborhood more attractive, these sorting effects are negligible at the scale examined in the policy counterfactuals. I demonstrate this with a partial-equilibrium calculation for each policy counterfactual that comprises my main results by holding counterfactual prices fixed and recomputing each household’s location choice, including new income from capital gains distributed to the owner share occupying each neighborhood, calculated from the ACS sample. I find that this procedure repeated over both main policy counterfactuals produce maximum re-sorting effects equivalent to 0.1% of an income groups population in any neighborhood. Further, because I hold prices fixed in this experiment, the small population changes it implies represent an upper bound on re-sorting due to capital gains.

3.6.1 Understatement of Price Effects for Low-Income Groups

The climate gentrification channel examined in this paper operates through rental prices, so the net effect that cooling investment alone has on low-income renting households depends on how much price changes driven by increases in neighborhood amenity value hurt them relative to the value they place on cooling. In this section I discuss how Cobb-Douglas preferences bias the inferred price burden for households across the income spectrum. Applying Roy's identity to the Stone-Geary indirect utility of Section 3.2 shows that the money-metric welfare loss from a marginal increase in the price of housing, expressed as a share of income, is equivalent to the household's housing expenditure share:

$$b_{wn} \equiv \frac{P_n^o h_{w,n}^*}{I_w} = \beta + (1 - \beta) \frac{P_n^o \bar{h}}{I_w}. \quad (12)$$

Because $\bar{h}$ is a fixed quantity of housing that does not scale with income, (12) falls with I_w . A rent increase costs a low-income household a larger share of its income than it costs a high-income household, and it does so for a reason that has nothing to do with preferences over neighborhoods. Setting $\bar{h} = 0$ recovers the homothetic Cobb-Douglas case used throughout the sorting literature, and the share collapses to β for every household at every income. A homothetic model records no variation in incidence across income: with $\bar{h} = 0$ the burden of a price increase is the same share of income for every household. In the calibrated model the Stone-Geary share falls from 0.43 at the lowest income to 0.14 at the highest, a gradient of 3.1, while the Cobb-Douglas share is constant at $\beta = 0.248$ by construction.¹⁰ Appendix G derives (12).

¹⁰There is an exception in this policy context. Where the minimum lot binds, the household is no longer on its demand curve, and it consumes the regulated lot rather than $h_{w,n}^*$. Its housing expenditure is R_n regardless of income. The share (12) is then replaced by $b_{wn} = R_n/I_w$, which falls with income even when preferences are homothetic. The minimum lot therefore functions similarly to a regulatory subsistence requirement, like $\bar{h}$, it requires a household to consume a fixed quantity of housing that does not scale with income, and it reproduces a small fraction of the regressive incidence that non-homothetic preferences generate on their own.

3.6.2 Overstatement of the Preference Gap for Cooling Between Low- and High-Income Households

Neighborhood amenity values are inverted from the model so that it reproduces how each income group sorts across neighborhoods. The amenity a_{wn} is therefore assigned whichever pattern of location choice the consumption value of the neighborhood cannot account for. Differentiating that inversion with respect to neighborhood temperature T_n splits the estimated value of cooling for group w into two parts,

$$\theta_w \equiv -\frac{\partial \log a_{wn}}{\partial T_n} = \underbrace{\zeta_w}_{\text{observed sorting on heat}} + \underbrace{c b_{wn}}_{\text{sorting explained by the budget}}, \quad c \equiv -\frac{\partial \log P_n^o}{\partial T_n} > 0. \quad (13)$$

The first term measures how strongly group w 's location choices move with temperature, setting aside what the budget can explain. It is the same in both demand systems, because both are calibrated to reproduce the same populations and so read the same sorting of income groups across temperature. The second term is the budget's contribution, and by (12) its size is set by the housing expenditure share.

Cooler neighborhoods are more expensive. A low-income household's high chance of living in a hot neighborhood is therefore partly explained by the cooler alternative being unaffordable, not by a taste for heat. How much of it is explained this way depends on how much of the household's income goes to housing, and the two demand systems disagree on that. Stone-Geary lets a household spending forty-three percent of its income on housing attribute much of its location to the budget. Cobb-Douglas holds the housing share fixed across income, so it cannot, and it concludes instead that low-income households live in hot neighborhoods because they mind the heat less, reading the difference as a low willingness to pay for cooling.

Taking the difference of (13) across income groups, the D_w terms cancel and the bias is

$$\left[\theta_H^{CD} - \theta_L^{CD} \right] - \left[\theta_H^{SG} - \theta_L^{SG} \right] = c \left[b_L - b_H \right] > 0, \quad (14)$$

the price-temperature gradient times the gap in housing expenditure shares, which is zero only when housing demand is truly homothetic.¹¹ Appendix G derives (13) and (27).

The bias matters for how the distributional benefits cooling policy is assessed. By widening the estimated preference gap, the homothetic model treats the concentration of low-income households in hot neighborhoods as a matter of how much each group values cooling, rather than the housing cost burden that drives it. A planner using it would read a cooling improvement in a poor neighborhood as worth less to residents than it is, and would understate the affordability pressure the improvement sets in motion, which the homothetic model already understates through its treatment of the price effect.

3.6.3 Overstatement of Welfare Gains from Zoning Reform

The welfare a household recovers when the minimum lot requirement is relaxed is the value of no longer being forced to consume more housing than it would choose, and this value grows as the wedge between the minimum lot and the household's optimal bundle widens. In this subsection I discuss how homothetic demand overstates this wedge, along with the efficiency gains it attributes to zoning reform for low-income households.

At baseline prices and under non-homothetic Stone-Geary demand, a household's desired housing consumption is $h_{wn}^* = \bar{h} + (\beta/P_n^o)(I_w - P_n^o\bar{h})$. The subsistence quantity $\bar{h}$ is the same for every household, absorbing a larger share of spending at lower incomes. This effect reproduces the income gradient in the housing expenditure share b_{wn} of (12), which falls from

¹¹The econometric estimates bear this out. Stone-Geary gives (0.082, 0.138) at the lowest and highest incomes and Cobb-Douglas gives (0.075, 0.151), so the homothetic specification widens the rich-poor gap in the estimated value of cooling from 0.056 to 0.076.

0.428 for the lowest income group to 0.137 for the highest in my data. Homothetic demand replaces this income-varying share with a constant $\beta_{CD} = 0.248$, the sample mean across income groups. A constant share understates the optimal housing expenditure share of low-income households, who choose 0.428, and overstates that of high-income households, who choose 0.137. Because the minimum lot binds almost entirely on lower-income households, understatement only occurs at the bottom of the income distribution. (Appendix J.1).

The value credited to zoning reform grows with the wedge between the regulated share R_n/I_w and the optimal share each model implies. By placing the low-income optimum too low, homothetic demand overstates the wedge, along with the gain it attributes to reform for each constrained household. Additionally, because Coobb-Douglas preferences put the low-income optimum below the true one, it treats more low-income households as bound by the minimum lot than are in fact bound, so it credits zoning reform to a larger share of them (Appendix J.1). Homothetic demand therefore overstates both how much each constrained household gains from reform and how many are constrained.

3.7 Housing Supply

Constant returns to scale in housing production imply that landowners of any parcel size choose the same capital intensity, which lets me express housing supply per parcel. Supply per parcel is tractable for this paper because minimum lot policies regulate density. On a parcel in neighborhood n , an absentee landowner faces a neighborhood-specific Cobb-Douglas technology and chooses housing supply by optimally choosing capital, produced from the numeraire at a rate of transformation of unity; with fixed land input. Following Macek (2024), I normalize housing supply by land area, yielding a per-unit-land supply of housing services in zone o of

$$A_n^o = \lambda_n \left(P_n^o \right)^{\varepsilon_n}, \quad (15)$$

where P_n^o is the price of a unit of housing services in zone o , $\varepsilon_n > 0$ is the floor-space supply elasticity in neighborhood n , and $\lambda_n > 0$ is an exogenous supply shifter. Under Cobb-Douglas production, land values are a constant fraction $1/(1 + \varepsilon_n)$ of gross housing revenue $P_n^o A_n^o$. Landowners in the regulated zone are subject to a minimum lot size in acres, ℓ_n , so that housing supplied per minimum lot is

$$A_n^{\text{reg}} \ell_n = \lambda_n (P_n^{\text{reg}})^{\varepsilon_n} \ell_n. \quad (16)$$

The minimum-lot rental cost is then

$$R_n \equiv P_n^{\text{reg}} A_n^{\text{reg}} \ell_n, \quad (17)$$

the expenditure required to rent the mandated bundle of housing services on a regulated lot. The unregulated has no minimum lot requirement, which I represent as $\ell_n = 0$.

4 Data and Calibration

This section describes how the model is taken to the data. Each parameter is either estimated from observed data, calibrated to match an equilibrium moment, or taken from the literature; I describe this process throughout. The calibrated model is then used to simulate the counterfactual cooling and zoning policies of Section 5.

4.1 Sample and Income Groups

Population, income, age, and tenure come from the 2015–2019 American Community Survey (ACS) five-year estimates at the block-group level, accessed through IPUMS NHGIS (Manson et al., 2022; U.S. Census Bureau, 2020). I partition households into four groups

defined against the HUD area median family income (AMI) for a four-person household in the Phoenix–Mesa–Scottsdale metropolitan area, \$66,200 in FY2017 (U.S. Department of Housing and Urban Development, 2017), rather than against sample quantiles, so that the boundaries correspond to program-relevant levels of housing need.¹² The groups, expressed in 2025 dollars, are:

Group	AMI band	Income range (2025 \$)	Mean income	Share of sample
Low	below 50%	< 44,100	\$34,793	25.1%
Lower-middle	50–80%	44,100–75,600	\$59,529	20.8%
Upper-middle	80–150%	75,600–126,000	\$99,388	23.5%
High	above 150%	> 126,000	\$209,273	30.6%

I apply five sample restrictions, described in full in Appendix B, that leave a final sample of 1,595 block groups. I drop block groups whose age composition identifies them as student or retirement communities, where more than 75 percent of residents are aged 18–24 or 65 and over, since residence there is tied to an institution rather than the neighborhood choice the model represents and reported money income poorly measures resources. I drop vacation-home enclaves, where seasonally vacant units exceed 90 percent of the vacant stock. I drop the exurban fringe, the top one percent of block groups by land area and by distance to the downtown core, residents in these areas do not plausibly sort in response to the margins that my model represents. I drop the top decile of block groups by minimum-lot cost, ultra-luxury enclaves in which the true income of high-group residents lies above the capped top income bracket, so that recorded income understates the resources of the households actually sorting there.¹³ Homeownership rates ϕ_{wn} are the owner-occupied share by block group and income

¹²I anchor to the four-person AMI so the thresholds are not set so low as to misclassify larger low-income families, and I inflate the FY2017 thresholds to 2019 dollars before assigning ACS income brackets. Each group’s representative income is its population-weighted mean bracket midpoint, with the open-ended top bracket assigned \$250,000; for the low group the mean excludes households below \$20,000 in 2019 dollars, below which reported money income does not reliably reflect resources.

¹³Additionally, I drop block groups with fewer than two income groups present or fewer than 25 households.

from ACS tenure tabulations. The regulated housing share is the single-family detached share of units, taken from the census structure tabulation (ACS B25024) rather than from transaction counts, because multifamily properties often transact as whole buildings and would distort a transaction-based structure share.

4.2 Housing Prices

Each block group has a regulated price P_n^{reg} for the single-family zone and an unregulated price P_n^{unreg} for the multifamily zone. I recover P_n^{reg} from a hedonic regression of log sale prices on structural characteristics, sale-year and sale-month effects, and a block-group fixed effect, estimated on arms-length residential sales from the Maricopa County Assessor for 2012–2019:

$$\ln p_{i,n,t} = \gamma \text{sf}_i + X_i' \eta + \delta_{\text{year}(t)} + \delta_{\text{month}(t)} + \theta_n + \varepsilon_{i,n,t}, \quad (18)$$

where sf_i indicates a single-family structure and X_i collects living area, lot area, stories, bathrooms, age, and pool, patio, garage, and roof-type indicators. I set $P_n^{\text{reg}} = \exp(\hat{\theta}_n)$, normalized to a mean of one, so the index measures the block group’s price of housing net of its structure mix. Because most transacting units are single family and multifamily buildings sell whole rather than per unit, the data do not identify a separate unregulated price directly. I therefore hold P_n^{reg} fixed and calibrate P_n^{unreg} from the condition that the model’s regulated population share equals the observed regulated unit share; the minimum lot lowers the value of the regulated option, so P_n^{unreg} rises above P_n^{reg} by an amount increasing in the regulatory distortion, and equals it absent a binding lot. Appendix C gives the clearing condition and the solution algorithm.

This follows from the variation the model requires to calibrate parameters, a block group with only one income group does not offer within-neighborhood income variation for the sorting model to fit, while neighborhood with too few households yields income shares that are too noisy to measure reliably.

4.3 Preferences

The Stone-Geary parameters $\bar{h}$ and β jointly determine each group's housing expenditure share and are calibrated to match observed shares by income. Using the 2015–2019 ACS microdata for Maricopa County, I compute the mean ratio of housing cost to income for each group and choose $(\bar{h}, \beta)$, given the group incomes and calibrated prices, to reproduce them. The target shares are 0.415, 0.290, 0.204, and 0.131 from low to high income, which the calibration matches with $\beta = 0.077$ and $\bar{h}$ equal to \$13,263 in annual subsistence housing expenditure.

4.4 Neighborhood Amenities and the Value of Cooling

I recover the group-specific amenity a_{wn} by inverting the sorting model of Section 3.4. Given calibrated prices, incomes, and preferences, a_{wn} is the vector of neighborhood values that reproduces the observed population share of each income group in each block group. The inversion identifies a_{wn} up to a normalization, and removes endogenous price and consumption terms analytically rather than through an instrument.

To recover how urban heat enters the amenity, I estimate the marginal value of cooling θ_w by two-stage least squares of log amenity on block-group nighttime temperature, separately by income group:

$$\text{Temp}_n = \pi_0 + \pi_1 \Gamma_n + X'_n \pi + \psi_{\text{zip}(n)} + \lambda_{\text{puma}(n)} H_n + \nu_n, \quad (19)$$

$$\log a_{wn} = \tau_0 - \theta_w \widehat{\text{Temp}}_n + X'_n \tau + \psi_{\text{zip}(n)} + \lambda_{\text{puma}(n)} H_n + \varepsilon_{wn}, \quad (20)$$

where Temp_n is the mean July nighttime near-surface air temperature from the NCAR HUMID dataset at one-kilometer resolution, and $\theta_w > 0$ is the marginal willingness to pay for one degree of cooling. Temperature is endogenous because it is correlated with unobserved

built-form amenities. For example, hotter block groups are denser and less vegetated, features that households value directly. I instrument temperature with the gradient alignment score Γ_n , which exploits nighttime heat advection. Γ_n measures how closely the prevailing near-surface airflow into a block group, simulated with NOAA’s HYSPLIT trajectory model, aligns with hotter- or cooler-than-average land just outside the block group’s own boundary. It shifts temperature through air transported from neighboring land while remaining orthogonal to the block group’s own built environment, isolating exogeneous variation in heat Appendix D reports the full instrument construction.

The specification absorbs two sets of fixed effects. The terms $\psi_{\text{zip}(n)}$ are zip-code fixed effects that absorb spatially persistent amenities such as school quality. The term $\lambda_{\text{puma}(n)} H_n$, where $H_n = \mathbb{I}[\text{elevation}_n \geq p_{90}]$ indicates a block group in the top decile of elevation, is a PUMA-specific intercept for high-elevation block groups; it absorbs the distinct amenity premium of foothill and mountain-view terrain, which is both cooler at night and separately valued, and which varies across sub-regions of the metropolitan area.¹⁴

The estimates above inform how a temperature change enters welfare; the size of the temperature change the counterfactuals apply is itself constructed from the Cool Corridors planting design rather than assumed. I convert the program’s stated planting density goal and species mix into the canopy area each planned, scheduled and completed corridor adds (the first corridor was completed in 2021, after the household data calibration period that ended in 2019), allocate that canopy to the block groups it cools weighted by percent area overlap, and map the resulting increase in block-group canopy cover into an air-temperature reduction using literature estimates derived from neighborhood-scale ENVI-met simulations

¹⁴The controls X_n are exogenous physical and locational features: elevation and distance to a major road, both from the USGS; distance to the downtown core; parcel greenspace from the Central Arizona–Phoenix LTER vegetation data; the distance to the center of the residential subdivision, a proxy for proximity to the parks, pools, and community facilities typically clustered at the center of suburban master-planned developments; and an indicator for the north- and central-Mesa tracts that concentrate RV parks and faith-based living communities, whose residents choose housing on constraints unlike the rest of the sample. The instrument is strong, with first-stage F -statistics between 56 and 63 across the four groups.

for a typical Phoenix residential neighborhood (Middel et al., 2014). Applying the estimated θ_w to this reduction gives the amenity increase each treated block group receives. Appendix F details each step, from trees planted to canopy cover to degrees of cooling.

4.5 Minimum Lot Size and Cost

I measure the minimum lot requirement ℓ_n from observed zoning rather than from parcel sizes alone. For Phoenix and its eight largest surrounding municipalities, I obtain each city’s residential zoning GIS layer and map every zoning code to the minimum lot size published by that city, assigning zero to multifamily, townhome, and manufactured-housing districts, which impose no single-family lot floor and constitute the model’s unregulated zone. Each block group is assigned the minimum lot of the zoning district covering the largest share of its area. For less-populated municipalities without a coded layer, and for unincorporated county land, I fall back to the modal single-family parcel size from Assessor sales, and where a large-lot rural or environmentally-constrained district covers a block group whose developed stock sits on much smaller lots, I use the observed modal lot instead. Appendix E gives the crosswalk and the reconciliation rule.

The minimum-lot cost R_n is the annual cost of renting the mandated lot, which by Equation (17) is the annualized housing value per acre times the lot size, $R_n = (P_n^{\text{reg}} A_n^{\text{reg}}) \ell_n$. I construct housing value per acre as total single-family sale value divided by single-family land area within the block group, using arms-length sales from 2012 on, with each pre-2019 sale lifted to the 2025 market by Phoenix real house-price appreciation between its sale year and 2019, measured by the FHFA house-price index net of CPI (Federal Housing Finance Agency, 2024), and annualized at a five percent user cost of land.

4.6 Supply Elasticities

The neighborhood floor-space supply elasticities ε_n in the developer’s problem are taken from the literature. I assign each block group the census-tract housing unit housing-supply elasticity estimated by Baum-Snow and Han (2024), so block groups in the same tract share an elasticity.

5 Counterfactual Design

I evaluate the cooling and zoning policies against the same benchmark, which is informed by the income, population, zoning, and housing price data discussed in Section 4. Additionally, I update the benchmark to include the small zoning reform response that Arizona’s House Bill 2721, the Middle Housing Bill passed in 2024 and effective January 2026 implies for the one mile radius surrounding Phoenix’s downtown. This includes relaxing the minimum lot in this area to a quarter of its current level in every block group intersecting the one-mile radius of the downtown core, and re-solving for equilibrium. This area represents a negligible amount of overlap with the total city area that comprises Cool Corridors and is therefore mainly included in the benchmark for good measure. All reported changes are measured against this HB 2721 baseline.

The cooling shock is applied to the 179 block groups that overlap a Phoenix Cool Corridor, the program’s actual planting footprint. I raise the amenity of each treated block group by the amount that the estimated marginal value of cooling θ_w implies for the temperature reduction the corridor’s planting delivers to that block group. I construct that temperature reduction from the program’s own planting density and species mix, rather than from an assumed canopy target; Appendix F describes how I arrive from program trees planted to canopy cover to neighborhood air-temperature reduction.

I report two main counterfactuals, corresponding to the two panels of Table 3. In *cooling alone*, I apply the cooling shock to the 179 corridor block groups and leave the minimum lot at its baseline level. In *cooling with targeted zoning reform*, I apply the same shock and additionally relax the minimum lot to a quarter of its current level in the highest-ranked corridor block groups. Table 4 varies the number of block groups reformed; the main specification reforms forty, the point beyond which the low-income welfare gain is essentially flat.

5.1 Selecting Zoning Reform Location

I target block groups for zoning reform by ranking on the improvement in the block group itself, which is determined by the change in the value of its option as a location to reside in once the minimum lot is relaxed. For a group- w household this is

$$\Delta v_{wn} = \frac{I_w}{\rho_{\text{within}}} \left[\log \Omega_{wn}^{\text{cool}+\text{zone}} - \log \Omega_{wn}^{\text{cool}} \right], \quad (21)$$

where Ω_{wn} is the within-block-group inclusive value of Equation (8), and the two superscripts denote the cooling-plus-zoning and cooling-only equilibria. Because the amenity is unchanged by upzoning, it cancels from the difference, so Δv_{wn} isolates the value of relaxing the constraint, evaluated in general equilibrium after prices adjust. I rank corridor block groups on the low-income measure $\Delta v_{\text{low},n}$, weighted by the block group's low-income population, and select the top N .

I rank on $\Delta v_{\text{low},n}$ because this is the appropriate measure for a non-marginal reform. A shadow price on the minimum lot is a sufficient statistic for welfare only under a marginal change (Chetty, 2009). Because the reform relaxes the lot size discretely, I rank on the general-equilibrium improvement instead of the constraint's multiplier. I find that the forty block groups, 22 percent of the block groups within the cool corridor neighborhoods, deliver

96 percent of the low-income welfare gain that reforming all of them would deliver (Table 4). Furthermore reforming only the first ten, carry the high-to-low welfare ratio from 27.5 to 1.88. I perform this routine to understand the minimum number of tracts that zoning must be reformed in to produce nearly all benefit to low income renters within Cool Corridor neighborhoods. This serves to help determine appropriate upzoning targets in combination with an improvement if the policy aim is to most benefit low income households.

6 Results

This section presents the empirical results in two parts. I first report econometric estimates of how neighborhood cooling capitalizes into amenity value for each income group, identified from the gradient alignment instrument of Section 4.4; these estimates supply the amenity shock used in the counterfactuals. I then report the welfare effects of my policy counterfactuals, decomposed by income group and by tenure, and I show how those effects depend on the choice of housing demand system.

6.1 Econometric Results

Table 1 reports two-stage least squares estimates of Equations (19) and (20), in which log neighborhood amenity value is regressed on mean July nighttime air temperature, instrumented with the gradient alignment score. The effect is negative and statistically significant for all four income groups: a one-degree Fahrenheit reduction in July nighttime temperature raises neighborhood amenity value by between 6.5 percent for lower-middle-income households and 13.8 percent for high-income households. First-stage F -statistics between 56 and 63 indicate a strong instrument. I take the negative of each coefficient as the income-group-specific marginal value of cooling θ_w that governs how the Cool Corridors improvement enters

utility in the counterfactuals.

Because the amenity is inverted from the sorting model, its estimated value depends on the assumed demand system. Table 2 repeats the estimation on amenities inverted under Cobb-Douglas preferences, for use in the demand-system comparison of Section 6.2.4. The two sets of estimates differ, with the Cobb-Douglas specification assigning a lower value of cooling to low-income households (7.5 against 8.2 percent) and a higher value to high-income households (15.1 against 13.8 percent). Thus, homothetic demand understates preferences for cooling for low income residents and overstates them for high income households. This widening of the rich-poor gap in the estimated value of cooling is the preference-channel bias that Section 6.2.4 traces to the demand system.

Table 1: IV: Effect of July nighttime air temperature on log neighborhood amenity value, Stone-Geary demand

	(1)	(2)	(3)	(4)
	Low inc.	Med-low inc.	Med-high inc.	High inc.
Block group July temp. °F	-0.082**	-0.065*	-0.073**	-0.138***
	(0.042)	(0.038)	(0.033)	(0.043)
First-stage F-stat	62.60	60.45	61.28	55.75
Observations	1,296	1,302	1,304	1,277

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Two-stage least squares estimates of the effect of mean July nighttime air temperature on log neighborhood amenity value, estimated separately by income group under Stone-Geary demand. Temperature is instrumented with the gradient alignment score (Appendix D); all specifications absorb zip-code fixed effects and control for elevation, distance to the downtown core, distance to a major road, and parcel greenspace. A negative coefficient means higher July heat lowers amenity value; the implied marginal value of one degree of cooling is the negative of the coefficient. These estimates supply the income-group cooling values used in the counterfactuals.

Table 2: IV: Effect of July nighttime air temperature on log neighborhood amenity value, Cobb-Douglas demand

	(1)	(2)	(3)	(4)
	Low inc.	Med-low inc.	Med-high inc.	High inc.
Block group July temp. °F	-0.075*	-0.065*	-0.079**	-0.151***
	(0.042)	(0.038)	(0.034)	(0.044)
First-stage F-stat	60.21	59.48	63.22	56.87
Observations	1,296	1,302	1,304	1,277

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: As in Table 1, but estimated on amenity values inverted under Cobb-Douglas demand. These estimates are used in the demand-system comparison of Section 6.2.4.

6.2 Counterfactual Results

I report the welfare effects of the two counterfactuals, both solved in general equilibrium against the HB 2721 policy baseline, in three parts. Throughout, I decompose each household’s equivalent variation into three components that sum to the total welfare effect exactly, the value of the cooling amenity, the value of removing the zoning distortion, and the net effect of the price change, and I report effects per household within the 179 Cool Corridor block groups, with citywide aggregates in Appendix H. Section 6.2.1 shows that targeted zoning reform reverses the regressive incidence of cooling for renters; Section 6.2.3 shows that corridor owner-occupiers gain under the combined policy regardless of how their asset position is treated; and Section 6.2.4 shows that a homothetic demand system, unable to reproduce the income gradient in housing budget shares, misstates each component of the welfare result relative to non-homothetic demand.

6.2.1 Renter outcomes: cooling alone versus cooling with targeted zoning reform

I evaluate two main policy counterfactuals, both against the HB 2721 policy baseline and both solved in general equilibrium, so that prices, housing quantities, and household locations adjust together. The first applies the Cool Corridors amenity improvement and leaves the minimum lot requirement unchanged. This is the climate policy as Phoenix has designed it, and it isolates what happens when a city improves a neighborhood with cooling infrastructure without changing what may be built there. The second applies the same cooling improvement and simultaneously relaxes the minimum lot to a quarter of its current level in forty Cool Corridor block groups. Those forty are the highest-ranked on Δv_{un} , the money-metric value to a low-income household of a block group becoming a cheaper place to live once the minimum lot no longer forces it to buy more housing than it wants, developed in Section 5.1.

Across all renter households in my Maricopa County sample, cooling alone delivers a net equivalent variation gain of \$34.9 million a year; pairing it with targeted zoning reform raises that to \$59.6 million. The \$24.7 million difference is the citywide contribution of the zoning reform, holding the cooling policy fixed, and it comes from two sources. The first is \$11.1 million in newly created value from removing the zoning distortion, the deadweight loss of the minimum lot returned to the households it bound, of which \$8.3 million accrues to the lowest income group alone. The second is a \$13.6 million improvement in the net price effect: under cooling alone the price adjustment imposes an aggregate welfare loss of \$5.8 million a year on renters, as rising demand for the cooled neighborhoods bids up rents, whereas under cooling paired with the reform the same channel delivers an aggregate gain of \$7.8 million, because relaxing the minimum lot expands the supply of housing units and moderates prices across the county. Appendix H reports the full accounting over the full sample. The remainder of this section reports the distributional effects of these counterfactuals per renter household, inside the Cool Corridor block groups (N=179), which is where both policies effects are most

concentrated.

Table 3 organizes the results for renter households residing in Cool Corridor neighborhoods post counterfactual shock into three blocks. *Cash* tabulates changes in what a household pays: rent paid is its total annual housing expenditure after each policy shock, and the housing cost burden is that expenditure as a share of income. *Welfare* is the equivalent variation, the annual sum a household would need at baseline prices to be as well off as the policy makes it, together with the three components into which it decomposes. *Sorting* is the change in the corridor population of each group.

The three welfare components are constructed sequentially and sum exactly to the total equivalent variation for that group in each counterfactual. *Cooling amenity* is what the improvement is worth to households at baseline minimum lots and baseline prices. *Value of removing the zoning distortion* measures the component of equivalent variation from relaxing the minimum lot constraint, at baseline prices. It measures what a household gains from being permitted to buy the housing it would choose rather than the housing the regulation requires. This value is created rather than transferred, and it is zero for households that the minimum lot never constrained. *Net price effect* lets prices adjust to counterfactual outcomes. It measures the full welfare consequence of the price change, and it is larger than the change in rent paid because the rent bill is measured after the household has already adjusted its housing. A household that responds to higher prices by renting less keeps its bill down, but only by giving up housing or other goods it valued, and that foregone consumption is a welfare loss the adjusted rent change does not capture. The net price effect in Table 3 counts both: the higher price paid on the housing that the household keeps, and the value of the housing or other consumption it gave up. For this reason the cash block and the welfare block do not necessarily move together.

Panel A shows that the climate policy on its own is regressive in the neighborhoods it targets. Low-income renters place substantial value on the cooling amenity but keep almost

none of it once housing prices adjust. The amenity is worth \$432 a year to a low-income household and \$4,281 to a high-income one (1.2% and 2% of income respectively), yet after prices adjust the low-income household is left with \$20 and the high-income household with \$3,331 in total equivalent variation. As a share of income, the high-income gain is twenty-seven times the low-income gain. Prices rise because demand for Cool Corridor block groups rises, and the new demand is disproportionately high-income: the high-income corridor population grows 5.73 percent while the low income population in the cool corridors neighborhoods decreases by 0.21%. Thus, the net price effect absorbs \$412 of the \$432 the improvement is worth to a low-income household, or 95%, against \$950 of the \$4,281 it is worth to a high-income household, or 22%.

The cash block shows the same pattern in terms of adjusted housing costs. A low-income household pays \$277 more in annual rent and a high-income household pays \$391 more, a difference of less than half, while the amenity value they receive differs by a factor of ten. The rent bill, however, understates the harm, because it is measured after households have already adjusted their housing. The net price effect is larger than the increase in rent paid: a household that responds to higher prices by renting less keeps its bill down, but only by giving up housing it valued, and that foregone housing is a welfare loss the bill does not record. Low-income households can least afford this response and are also least able to make it. The minimum lot binds on 33.4 percent of them in the corridor neighborhoods (Appendix I), so they cannot shrink their housing and instead absorb the full price increase by cutting non-housing consumption. The sorting row confirms that the Cool Corridors policy raises the housing burden on low-income households already living in these neighborhoods rather than displacing them: the low-income corridor population falls by only 0.21 percent. Low-income households do not leave the cooled neighborhoods. They stay, and their gains from the improved amenity are almost entirely offset by the net price effect.

Panel B shows that pairing the Cool Corridors policy with targeted zoning reform reverses

the regressive result of the cooling-only counterfactual, mainly by producing large gains for low-income renters by relieving the distortion of the binding minimum lot. The low-income gain rises from \$20 to \$506 a year while the high-income gain rises only from \$3,331 to \$3,395, and the ratio of the two as a share of income falls from 27.4 to 1.11. The source is the value of removing the zoning distortion: \$444 of the \$506 a low-income household gains, and none of the \$3,395 a high-income household gains. The low-income household's annual rent bill falls by \$210, because the relaxed minimum lot allows it to downsize, and the drop in quantity more than offsets a higher price per unit.

Under the combined policy counterfactual, low income welfare, however, still registers a net price effect loss of \$369, close to the \$412 under cooling alone: the cooling drives up demand and the price per unit rises regardless of the reform. Furthermore, while not optimal, the housing the household forgoes in downsizing was housing it valued, even though the minimum lot had forced it above its preferred level. The net price effect accounts for this loss. The low income household's large welfare gain relative to Cool Corridors alone therefore comes from the \$444 value of choosing its desired housing bundle, which more than offsets the price effect it continues to bear. Because the minimum lot binds disproportionately on low-income corridor households, relaxing it returns value primarily to poorer households. The reform therefore does not redistribute the cooling benefit from rich renters to poor renters; it creates new value where the constraint was binding.

The result is progressive in rent payments as well. A low-income household pays \$13,510 a year in rent, \$210 less than it paid before any policy was enacted, and its housing cost burden falls from 39.4 to 38.8 percent, while the two highest income groups continue to pay the cooling premium in full, at \$98 and \$366 above baseline. Greater affordability also draws low-income households into the improved neighborhoods: the low-income corridor population rises 4.22 percent, so more of the intended beneficiaries of the Cool Corridors program live in the cooled block groups than when cooling is not paired with zoning, and

each household pays less.

6.2.2 Sensitivity of main counterfactual results to spatial extent of zoning reform

The results in table 4 show that returns to zoning reform in the Cool Corridor neighborhoods diminish quickly, relieving policymakers the burden of reforming zoning over large areas. Relaxing the minimum lot in forty block groups raises the low-income gain in the combined Cooling and zoning counterfactual to \$506 a year against \$526 if all neighborhoods with binding constraints were upzoned. Thus, upzoning policy in forty block groups deliver 96 percent of what a corridor-wide reform would deliver. Table 4 holds the tree canopy in place and varies the number of Cool Corridor block groups in which the minimum lot is relaxed. Block groups enter each counterfactual in descending order of Δv_{wn} , the dollar value to a low-income household of reducing the minimum lot constraint in that block group. Because returns to low income renters diminish significantly after the counterfactual run with the most costly forty block groups constrained, I take forty as the main counterfactual: it is the smallest reform that captures essentially the whole gain, minimizing the administrative burden of passing such reform while achieving the majority of the benefit.

6.2.3 Owner-occupiers

Every household in the model solves the same problem and sorts across block groups as a renter. Tenure enters afterward, by crediting an owner-occupier with the change in the value of the home it owned prior to the policy shock. That change, reported in the first row of each panel of Table 5, is the change in the annual rental value of that home: a price effect applied to the fixed quantity of housing owned prior to the policy, since an owner cannot buy less of a home it already holds. It is distinct from the changes in rent paid and in amenity

value reported for renters in Table 3, because a renter’s rent bill responds to both the price and the quantity it chooses and can therefore fall when prices rise, while an owner’s home is a fixed quantity of housing whose value can only follow the price.

The value of the counterfactual changes to an owner-occupier in the Cool Corridor neighborhoods is depends on what it does with that change. I report four potential scenarios for owners, which differ only in which pieces of the household problem each is credited with. An owner who *stays and does not refinance* receives the cooling amenity and nothing else: it neither pays the new price on its housing, since it already owns it, nor realizes the change in its home’s value. An owner who *sells and leaves the metro* receives the change in home value and nothing else: it forfeits the cooling amenity, the new prices, and the relaxed minimum lot. An owner who *borrow against the home and stays* receives both the cooling amenity and the change in home value, but does not reoptimize its housing and so gains nothing from the relaxed minimum lot. An owner who *sells and reoptimizes its housing* is credited with the full household problem, including the value of removing the zoning distortion and the net price effect, plus the change in the value of the home it sold. The four bracket every intermediate case, and only the last is credited with the gain from zoning reform.

Under both of my primary counterfactuals, which are represented by Panels A and B in 5, corridor owner-occupiers gain under every assumption. In Panel A, which tabulates the effects from Cool Corridors alone, the cooling raises the annual value of a low-income owner’s home by \$410 and a high-income owner’s by \$915, and it is worth \$432 and \$4,281 respectively to those who stay and consume it. Adding targeted zoning reform reduces the appreciation by roughly \$60 in every group, since supply expands where the minimum lot is relaxed and prices there fall, but it does not reverse the positive gains to homeowners from the neighborhood amenity improvement: the cooling raises prices by far more than the reform lowers them. For an owner who reoptimizes, the reform raises the low-income gain from \$430 to \$856, because the value of removing the zoning distortion more than replaces the

forgone appreciation, while leaving the high-income owner essentially unchanged at \$4,250 against \$4,246.

Panel C isolates zoning reform from the cooling, and shows why upzoning on its own without any amenity improvement is politically difficult. In the counterfactual where only zoning reform is applied with no cooling improvement, An incumbent owner who sells loses \$60 a year, or \$130 in the forty block groups actually targeted, which capitalizes at a rate of 5% into a decline of roughly \$2,600 in the sale price of a low-income owner's home. Per owner that loss is regressive. But only 36.3 percent of low-income corridor households own anything, against 80.6 percent of high-income households, so the loss per household of the group is flat across the income distribution and smallest at the bottom, at \$47 against \$53. Panel B then shows that the losses to homeowners are singular to zoning, and they do not survive the policy I actually evaluate: when bundled with the cooling amenity, the same zoning reform leaves each group of owner-occupiers in the Cool Corridor neighborhoods better off, because the amenity raises home values by more than the upzoning lowers them.

6.2.4 Non-homothetic versus homothetic housing demand

The results discussed above are generated from a non-homothetic demand system, in which the share of income a household spends on housing falls as income rises. The environmental sorting literature more commonly assumes homothetic, Cobb-Douglas preferences, under which expenditure share on housing is constant. This section re-solves the cooling counterfactual under homothetic demand and shows that the choice of demand system matters not only for the measured size of cooling's regressive incidence, but for what the regressive incidence is attributed to. Panel A of Table 6 reports that while both demand systems produce regressive results, they disagree on whether regressivity is attributed to differences in preferences for cool neighborhoods or differences in price impacts across income groups. Under the non-homothetic housing demand, the regressivity is roughly half (52%) a matter

of preference, that high-income households value cooling more, and half (48%) a matter of affordability, that the same rent increase consumes a larger share of a low-income budget. Under the homothetic system, regressivity gets attributed almost entirely (87%) to preference. Because homothetic demand holds the housing budget share constant across income, it cannot represent the affordability channel through which climate gentrification operates, and a planner relying on it would read the regressive incidence of cooling as a feature of preference rather than housing-market pressure that policy can relieve.

Table 6 reports the comparison across the income distribution. Under non-homothetic Stone-Geary demand, the model reproduces observed household expenditure shares on housing. The share a household would choose at its unconstrained optimum falls from 0.428 for the lowest income group to 0.137 for the highest. Under homothetic Cobb-Douglas demand every income group spends the same fraction, fixed at the sample mean of $\beta = 0.248$. Because a single value of β cannot reproduce housing shares that vary across the income distribution, the homothetic model cannot reflect that housing is a necessity absorbing a larger part of a poor household's budget than that of a higher income household. All following differences between the two models are generated from this discrepancy between the two systems.

Panel B of Table 6 shows that the value each group places on the change in the cooling amenity is understated for low income households and overstated for high income households under non-homothetic preferences. Because the amenity is inverted from observed location choices, a model that cannot attribute the concentration of low-income households in hot neighborhoods to their lower budgets attributes it to preference instead. Therefore the Cobb-Douglas demand system interprets differences in observed location choices between low and high income households as a result of low income households caring less about heat and high-income households as caring more, widening the estimated preference gap between them. Cobb Douglas evaluates low-income cooling gain at \$395 versus \$432 under Stone-Geary, and the high-income gain at \$4,692 compared to \$4,281.

In contrast to the amenity effect, homothetic preferences understate the negative impact of neighborhood price increases to low income groups and overstate the negative impact to wealthy households. Because the burden of a marginal rent increase is the housing expenditure share under both systems, the net price effect inherits the full income gradient on which the two demand systems disagree, so this effect diverges more between the poorest and wealthiest households than the amenity effect. Under Stone-Geary demand the welfare cost of the cooling-induced rent increase is steadily regressive, falling from 1.18 percent of income for the lowest income group (\$412) to 0.45 percent for the highest (\$950). Under Cobb-Douglas it is almost flat, 0.67 percent from the lower-middle group upward (\$400 to \$1,402) and rising only slightly at the bottom, to 0.83 percent (\$289). That the homothetic model carries this small regressive step is due to the minimum lot. Where it binds, which in the corridor neighborhoods is almost entirely on lower-income households (Appendix J.1), the household is forced onto the regulated quantity regardless of its preferences, so its housing share, and the burden a rent increase places on it, is pushed above β . But the lot is a weak substitute for non-homotheticity. It lifts the low-income housing share only to about 0.31 on average, short of the observed 0.42 expenditure share that low-income households spend, because Cobb-Douglas fixes the low income group's optimal share at the population mean $\beta = 0.25$ and binding lots closes only part of that gap. In the Stone-Geary results, housing is modeled as a necessity. Thus, every low-income household spends 0.42 of its income on housing whether or not a lot binds. Cobb-Douglas therefore still meaningfully understates the low-income price burden, \$289 versus \$412.

These differences distort the decision making process a planner may make when determining whether or not to implement a complementary housing policy to mitigate gentrification such a zoning reform. Because the homothetic model attributes almost all of cooling's regressivity to preference rather than affordability, it presents a smaller affordability problem to correct, and with it a weaker rationale for pairing cooling with housing-market reform. This distortion operates before any reform is considered. However, were the targeted reform

of Section 6.2.1 were to be applied, under the Stone-Geary demand system it would reverse the regressive incidence of cooling, raising the low-income renter's annual gain from \$20 under cooling alone to \$506, due to the opportunity to consume a smaller housing bundle in neighborhoods where the minimum lot is binding, reducing the affordability impact of the improved neighborhood. Importantly, the homothetic model also distorts the gain from removing minimum lot requirements were it to be enacted. Homothetic demand overstates gain from removing zoning, valuing the relaxed lot at \$1,156 for a low-income household compared to \$444 under Stone-Geary (Appendix J.2). This is because many more low income households appear to be constrained under homothetic demand, because the model assumes low-income households would choose to spend only 25% of their income on housing if they were free to do so, versus 42% in the non-homothetic model. This bias has clear implications for housing policy design. Because the homothetic model categorizes more than twice as many low-income households as constrained than the non-homothetic demand model does (Appendix J.1), it implies returns to upzoning that do not exist under non-homothetic model. Table 13 shows that when the first forty-block groups are reformed, Stone-Geary has realized 96 percent of its maximum low-income gain, while Cobb-Douglas has realized only 78 percent, and its gains keep rising through the largest reform considered. This would result in overly optimistic conclusions about the benefits that zoning reform provides low income renters with, if a complementary housing policy were to be considered.

This distortion is also policy relevant, because it counts more than twice as many low-income households as constrained (Appendix J.1), it implies returns to upzoning that continue beyond the point at which the non-homothetic model shows them exhausted, recommending reform across roughly twice as many block groups (Table 13) than would produce meaningful gains to low income renters.

Table 3: Renter outcomes: cooling alone versus cooling with targeted zoning reform

	Low < 44,100	Lower-middle 44,100–75,600	Upper-middle 75,600–126,000	High > 126,000
Panel A. Cooling alone				
<i>Cash</i>				
Rent paid (\$/year)	13,997	16,096	19,447	29,739
Change from baseline (\$)	+277	+315	+373	+391
Housing cost burden (%)	40.2	27.0	19.6	14.2
Change from baseline (p.p.)	+0.8	+0.5	+0.4	+0.2
<i>Welfare</i>				
Equivalent variation (\$/year)	20.17	107.35	522.51	3,330.87
Cooling amenity	432.10	584.43	1,155.31	4,280.97
Value of removing the zoning distortion	0.00	0.00	0.00	0.00
Net price effect (rent + foregone cons.)	−411.92	−477.08	−632.80	−950.10
Share of income (%)	0.058	0.180	0.526	1.592
<i>Sorting</i>				
Population change (%)	−0.21	+0.35	+1.66	+5.73
Household Pop. Change (levels)	−41	+45	+155	+269
Panel B. Cooling with targeted zoning reform				
<i>Cash</i>				
Rent paid (\$/year)	13,510	15,623	19,172	29,713
Change from baseline (\$)	−210	−159	+98	+366
Housing cost burden (%)	38.8	26.2	19.3	14.2
Change from baseline (p.p.)	−0.6	−0.3	+0.1	+0.2
<i>Welfare</i>				
Equivalent variation (\$/year)	506.46	377.93	645.17	3,394.63
Cooling amenity	432.10	584.43	1,155.31	4,280.97
Value of removing the zoning distortion	443.80	202.77	52.21	0.00
Net price effect (rent + foregone cons.)	−369.44	−409.27	−562.35	−886.34
Share of income (%)	1.456	0.635	0.649	1.622
<i>Sorting</i>				
Population change (%)	+4.22	+1.72	+1.97	+5.76
Household Pop. Change (levels)	+830	+222	+183	+270

Notes: Renter households in the 179 block groups that overlap a Cool Corridor, measured against the HB 2721 policy baseline. Income groups are bands of area median income, cut at 50, 80, and 150 percent of the HUD four-person median family income for the Phoenix-Mesa-Scottsdale area. The targeted reform relaxes the minimum lot to a quarter of its current level in the forty highest-ranked corridor block groups, selected by the criterion in Section 5.1; forty is the point at which the EV benefits to zoning for the low income group flatten, delivering 96 percent of what upzoning all Cool Corridor block groups would deliver. *Households entering (exiting)* applies the population change to the baseline corridor renter counts of Table 7; a positive value is net entry, a negative value net exit. The three welfare components sum exactly to the equivalent variation. *Cooling amenity* is what the Cool Corridors improvement is worth at baseline minimum lots and baseline prices. *Value of removing the zoning distortion* is what the household gains from being permitted to buy the housing it would choose rather than the housing the minimum lot requires, still at baseline prices. It is the deadweight loss of the regulation returned to the household on whom it was binding. *Net price effect* is the full welfare consequence of the price change; this exceeds the change in rent paid because it considers the price rise as well as decreases in housing consumed. For a low-income household under cooling alone, the rent ⁴⁷ rises \$277, but the household also purchases less housing where their lot constraint does not bind, so the total harm is \$412.

Table 4: Scaled zoning reform across Cool Corridor block groups

Block groups	Equivalent variation (\$/year)				Ratio High/Low	Low income		
	Low	L-mid	U-mid	High		Rent (\$)	Burden (%)	Pop. (%)
0	20.17	107.35	522.51	3,330.87	27.45	+277	40.2	−0.21
10	296.12	264.22	592.43	3,347.86	1.88	+73	39.6	+2.29
20	428.08	346.25	625.03	3,366.79	1.31	−65	39.2	+3.50
40	506.46	377.93	645.17	3,394.63	1.11	−210	38.8	+4.22
60	522.85	382.82	651.78	3,402.45	1.08	−253	38.7	+4.38
80	525.68	385.08	653.89	3,405.60	1.08	−262	38.7	+4.40
100	525.91	385.21	654.10	3,406.02	1.08	−263	38.7	+4.40

Notes: Renter households in the Cool Corridor block groups, measured against the HB 2721 policy baseline. The row for zero block groups is cooling alone. In every other row the Cool Corridors policy is implemented and the minimum lot is relaxed to a quarter of its current level in the listed number of block groups, chosen in descending order of the block-group-level improvement measure of Section 5.1. Ratio is the equivalent variation of the highest income group divided by that of the lowest, each expressed as a share of group income, so that a value above one is regressive and a value below one is progressive. The final three columns report, for low-income households, the change in annual rent paid, the level of the housing cost burden, and the change in the corridor population. Gains to low-income renters, for whom the minimum lot is most binding, flatten considerably after forty block groups are upzoned, implying that this reform is enough to relieve most of the renters in the Cool Corridors who faced a binding constraint. Thus, reforming the minimum lot in the top forty block groups delivers roughly 96 percent of the welfare gain and 80 percent of the rent reduction that reforming all 179 would deliver.

Table 5: Owner-occupier welfare in the Cool Corridors under four assumptions about owner behavior

	Low	Lower-middle	Upper-middle	High
Homeownership rate (%)	36.3	48.1	63.0	80.6
Panel A. Cooling alone				
Change in home value	410.07	469.10	620.23	914.76
<i>Equivalent variation, if the owner:</i>				
stays, does not refinance	432.10	584.43	1,155.31	4,280.97
sells and leaves the metro	410.07	469.10	620.23	914.76
borrowes against the home, stays	842.17	1,053.52	1,775.54	5,195.73
sells and re-optimizes housing	430.24	576.45	1,142.74	4,245.63
<i>Comparison:</i> renter equivalent variation	20.17	107.35	522.51	3,330.87
Panel B. Cooling with targeted zoning reform				
Change in home value	349.65	411.89	562.63	855.18
<i>Equivalent variation, if the owner:</i>				
stays, does not refinance	432.10	584.43	1,155.31	4,280.97
sells and leaves the metro	349.65	411.89	562.63	855.18
borrowes against the home, stays	781.75	996.32	1,717.94	5,136.15
sells and re-optimizes housing	856.10	789.82	1,207.81	4,249.81
<i>Comparison:</i> renter equivalent variation	506.46	377.93	645.17	3,394.63
Panel C. Zoning reform alone (isolating the effect on incumbent owners)				
Change in home value	-60.07	-55.54	-57.94	-58.04
<i>Equivalent variation, if the owner:</i>				
stays, does not refinance	0.00	0.00	0.00	0.00
sells and leaves the metro	-60.07	-55.54	-57.94	-58.04
borrowes against the home, stays	-60.07	-55.54	-57.94	-58.04
sells and re-optimizes housing	416.99	202.75	57.21	5.31
<i>Comparison:</i> renter equivalent variation	477.06	258.29	115.14	63.35
Subset: owners in the forty targeted block groups				
Change in home value, cooling with reform	358.71	323.81	487.79	729.99
Change in home value, reform alone	-130.00	-107.88	-101.52	-65.80
Reform alone, as a change in the sale price (\$)	-2,600	-2,158	-2,030	-1,316

Notes: Owner-occupying households in the 179 block groups overlapping a Cool Corridor, measured against the HB 2721 baseline. Entries are annual flows in dollars, except the change in the sale price, a stock obtained by capitalizing the annual flow at five percent. *Change in home value* is the change in the annual rental value of the dwelling the household owns. The homeownership rate is the population-weighted mean of the block-group-by-income ownership rate, over the corridor block groups. Equivalent variation differs across the four rows only through differences in assumed owner behavior. One who *stays and does not refinance* holds its home and mortgage fixed, so neither prices nor the relaxed minimum lot enter its budget: it receives the cooling amenity and nothing else. One who *sells and leaves the metro* realizes the change in home value and then consumes neither the cooling amenity, nor the new prices, nor the ability to buy housing under a lower minimum lot. One who *borrowes against the home and stays* receives the cooling amenity and extracts the change in home value without moving, but does not re-optimize its housing and so gets no relief from the minimum lot. One who *sells and re-optimizes its housing* liquidates its home, keeps the proceeds, and chooses housing freely at the new prices. Panel C isolates zoning reform from the cooling in order to measure the effect of upzoning on incumbent owners on its own, with no neighborhood amenity improvement.

Table 6: The cooling policy alone under non-homothetic (Stone-Geary) and homothetic (Cobb-Douglas) demand, across the income distribution

Panel A. Source of the rich–poor gap in total welfare (% of income)					
		Gap (p.p.)	Preference	Affordability	
Stone-Geary	SG	1.53	0.80 (52%)	0.73 (48%)	
Cobb-Douglas	CD	1.27	1.11 (87%)	0.16 (13%)	
Household income (2025 \$)		Low < 44,100	Lower-middle 44,100–75,600	Upper-middle 75,600–126,000	High > 126,000
Panel B. Welfare of the cooling policy, \$/year (% of income)					
Value of cooling	SG	432 (1.24)	584 (0.98)	1,155 (1.16)	4,281 (2.05)
	CD	395 (1.14)	584 (0.98)	1,251 (1.26)	4,692 (2.24)
Net price effect	SG	−412 (−1.18)	−477 (−0.80)	−633 (−0.64)	−950 (−0.45)
	CD	−289 (−0.83)	−400 (−0.67)	−671 (−0.68)	−1,402 (−0.67)
Total welfare (EV)	SG	20 (0.06)	107 (0.18)	523 (0.53)	3,331 (1.59)
	CD	106 (0.30)	184 (0.31)	580 (0.58)	3,290 (1.57)
Panel C. Housing cost burden (rent as % of income)					
Baseline	SG	39.4	26.5	19.2	14.0
	CD	30.7	26.5	24.9	24.8
With cooling	SG	40.2	27.0	19.6	14.2
	CD	30.8	26.6	24.9	24.8
Change (p.p.)	SG	+0.80	+0.50	+0.40	+0.20
	CD	+0.16	+0.13	+0.03	+0.00
Panel D. Sorting: change in corridor population (%)					
	SG	−0.21	+0.35	+1.66	+5.73
	CD	+0.73	+0.84	+1.86	+5.58

Notes. Key policy outcomes for renter households in the 179 Cool Corridor block groups measured under the cooling policy alone. Both models are solved on the identical cooling shock and calibrated to the same baseline populations, differing only in the demand system ($\bar{h} > 0$ under Stone-Geary, $\bar{h} = 0$ under Cobb-Douglas) and the income-specific value of cooling each recovers. Panel A decomposes the gap between the high- and low-income total welfare gain (as a share of income) into the part attributable to differences in the value each group places on cooling (preference) and the portion attributable to differences in how a rent increase burdens each group (affordability); under Stone-Geary the gap splits roughly evenly, whereas under Cobb-Douglas it is attributed almost entirely to preference, because the homothetic system cannot represent the income gradient in housing expenditure shares. Panel B reports each household’s equivalent variation and its decomposition into the value of the cooling amenity and the net price effect (which sum to the total), with the same quantity as a share of group income in parentheses. Panel C reports the housing cost burden, rent paid as a share of income. Panel D reports the change in the corridor population of each group.

7 Conclusion

Cities are investing heavily to cool their poorest, hottest neighborhoods, yet the same improvement that reduces heat exposure raises the rents that follow it, and those increases fall on the low-income residents the investment is meant to protect. This paper asks whether zoning reform can prevent that outcome, using a non-homothetic spatial equilibrium model of household sorting in Maricopa County, calibrated to Phoenix's Calibrated to Phoenix's Cool Corridors program, funded at roughly \$1.4 million annually. The central finding is that cooling on its own is regressive, raising low-income renters' housing costs while delivering them negligible benefits, despite their placing a large value on the cooling amenity itself. I show that pairing cooling with a targeted relaxation of minimum-lot zoning reverses this outcome. The zoning reform removes regulation that forced low-income households to consume more housing than they wanted, so that as prices rise they can re-optimize to their smaller, desired housing consumption. Their annual rent falls rather than rises, and their annual gain under the combined cooling and up-zoning scenario increases from \$20 to \$506.

When cooling is implemented alone, the Cool Corridors amenity is worth \$432 a year to a low-income renter, but once prices adjust the household keeps only \$20, against \$3,331 for a high-income renter, a gain twenty-seven times larger as a share of income. Low-income households largely do not leave, but instead stay and bear rent increases. Because the minimum lot binds on a third of these households, they cannot reduce their housing as prices rise, absorbing the increase by cutting other consumption instead. So, the cooling program alone raises an amenity that low-income households value but pushes prices up, absorbing almost all of its benefit.

I find that the combined policy is progressive, in contrast to the cooling-only counterfactual. The ratio of high- to low-income gains, measured as a share of income, falls from twenty-seven to roughly one when residential zoning is reformed. Furthermore, the combined policy

allows more of the program’s intended beneficiaries to live in the cooled neighborhoods, with the low-income corridor population rising by 4.22 percent. Further, annual rent for these households falls \$210 below its pre-cooling level. The reform does not redistribute the cooling benefit from high-income renters to low-income ones; it instead creates \$444 of new value for the average low-income household in cool corridor neighborhoods by allowing it to consume the housing it would choose rather than the housing the regulation requires.

My results also address a common objection to upzoning. Opposition to zoning reform is concentrated among homeowners who fear for their property values. I find that when zoning reform is applied on its own, home values fall slightly, reducing the sale value of a low-income owner’s home by roughly \$2,600. When zoning reform is paired with an amenity improvement that *increases* home prices, owner-occupiers in the upzoned neighborhoods are left better off, under any assumption about how the owner responds to the change, because cooling raises home values by more than the added density lowers them. Taken together, these results suggest that homeowners need not bear losses where a city pairs upzoning with a neighborhood improvement.

Zoning reform is one of several policies a city might use to protect low-income residents from rising rents, alongside housing vouchers and subsidies for low-income construction. However, the gains it delivers are from the recovery of a deadweight loss, as opposed to a transfer. The minimum lot forces low-income households to hold more housing than they would choose, and relaxing it returns the forgone value to them. The \$444 a low-income household gains is created rather than taken from another party, and it accrues to the households the constraint binds, who are disproportionately low-income, without large public expenditures. Where a cooling program already commits public funds, a policy that corrects an existing distortion without direct transfer costs is likely more politically feasible.

I further show that zoning reform need not be extensive. The gains to low-income renters are recovered quickly as it is scaled, upzoning only the forty most-constrained of the one

hundred and seventy-nine Cool Corridor block groups delivers 96 percent of what upzoning all Cool Corridor neighborhoods would. Cities may be able to protect low income residents against rent increases by focusing on a smaller footprint where minimum lots bind the most. This may limit the administrative and political costs of the reform to a fraction of the cooled area.

These conclusions rest on a non-homothetic demand system. I find that homothetic preferences, which are conventional in the sorting literature, attribute a much larger percentage of the regressivity associated with low income neighborhood improvement to differences in preferences across income groups, rather than to affordability consequences. Under both demand systems, the cooling policy enacted alone favors high-income renters. However, under the non-homothetic system that outcome is roughly half a matter of preference, because higher-income households value cooling more, and half a matter of affordability, because the same rent increase consumes a larger share of a low-income budget. Under the homothetic system, regressivity is attributed almost entirely to preference. This result stems from the constant housing expenditure shares that the homothetic model uses, which omits the affordability channel through which climate gentrification operates. A planner relying on it may interpret the regressive incidence of cooling as a matter of preference, which policy cannot change, rather than of affordability, which it can, and would find a weaker rationale for pairing cooling with housing-market reform. I find that were such a reform enacted, the homothetic model would then overstate its value, pricing the relaxed minimum lot at \$1,156 for a low-income household against \$444 under the non-homothetic model, and recommending reform across roughly twice as many neighborhoods as the low-income gain can support, because it incorrectly counts more than twice as many low-income households as constrained. The homothetic model thus errs in two directions: it understates the affordability harm that motivates the reform, and overstates the distortion the reform removes, so a planner is misled both about whether to reform and about how far to extend it. Recovering the incidence of policies aimed at low-income communities requires a demand system in which the housing

budget share falls with income.

Taken together, the results of this paper show that climate gentrification is not an unavoidable byproduct of cooling but a consequence of the supply environment in which cooling is delivered. Public climate adaptation spending and housing-supply policy are largely set through separate processes, and, treated separately, the first can undo its own distributional aim. However, applying them together lets the gains from cooling reach the residents it targets rather than capitalizing into their rents. The lessons from this policy context extend beyond urban heat to any amenity improvement made in a supply-constrained, low-income community. As local governments struggling with affordability crises scale up investment in their poorest neighborhoods, that investment protects the residents it is meant to help only when paired with the housing policy that governs who ultimately captures it.

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A Utility response to neighborhood cooling: full derivation

This appendix derives the equilibrium welfare change from a marginal improvement in a neighborhood’s microclimate, summarized in Section 3.3. A one-degree reduction in temperature in neighborhood n raises the amenity a_{wn} and delivers a direct utility benefit $\theta_w \equiv -\partial \log a_{wn} / \partial T_n \geq 0$. Because the cooler neighborhood becomes more attractive, households sort toward it and its equilibrium housing price rises. I write the induced change in the zone- o price as

$$\Delta P_n^o = \frac{\partial P_n^o}{\partial \log a_{wn}} \theta_w > 0.$$

Unregulated zone, or a non-binding regulated zone

Indirect utility is $V_{wn}^{\text{unreg}} = \log a_{wn} + k_w (I_w - P_n^{\text{unreg}} \bar{h}) / [P_n^{\text{unreg}}]^\beta$. Its derivative with respect to the price is

$$\frac{\partial V_{wn}^{\text{unreg}}}{\partial P_n^{\text{unreg}}} = k_w \left[-\frac{\bar{h}}{[P_n^{\text{unreg}}]^\beta} - \frac{\beta (I_w - P_n^{\text{unreg}} \bar{h})}{[P_n^{\text{unreg}}]^{\beta+1}} \right] < 0.$$

The equilibrium welfare change from one degree of cooling is the direct benefit plus the effect of the induced price change,

$$\Delta V_{\text{unreg},wn}^{\text{cool}} = \underbrace{\theta_w}_{\text{direct benefit}} - \underbrace{\Phi_{wn}}_{\text{rent penalty}}, \quad \Phi_{wn} \equiv -\frac{\partial V_{wn}^{\text{unreg}}}{\partial P_n^{\text{unreg}}} \Delta P_n^{\text{unreg}} \geq 0. \quad (22)$$

Applying Roy's identity, with $\partial V_{wn}^{\text{unreg}}/\partial I_w = k_w/[P_n^{\text{unreg}}]^\beta$,

$$-\frac{\partial V_{wn}^{\text{unreg}}/\partial P_n^{\text{unreg}}}{\partial V_{wn}^{\text{unreg}}/\partial I_w} = \bar{h} + \frac{\beta}{P_n^{\text{unreg}}} (I_w - P_n^{\text{unreg}} \bar{h}) = h_{wn}^*,$$

so in money-metric terms the rent penalty is $h_{wn}^* \Delta P_n^{\text{unreg}}$, and as a share of income it is the housing expenditure share b_{wn} of Equation (12) times the proportional price increase. The penalty therefore rises with b_{wn} , which falls with income, so a given cooling improvement costs a low-income household a larger share of its income. Cooling leaves a household worse off whenever $\Phi_{wn} > \theta_w$.

B Sample Construction and Restrictions

Population, income, age, tenure, structure type, and vacancy status are drawn from the 2015–2019 ACS five-year estimates at the block-group level via IPUMS NHGIS (Manson et al., 2022; U.S. Census Bureau, 2020): household counts by income (B19001), population by age (B01001), tenure (B25003), units in structure (B25024), and vacancy status (B25004). I assign each of the sixteen ACS income brackets its midpoint, with \$250,000 for the open-ended top bracket, inflate the HUD FY2017 four-person AMI thresholds of 50, 80, and 150 percent to 2019 dollars, and sort block-group households into the four groups. Each group's representative income is its population-weighted mean bracket midpoint; for the low group the mean excludes households below \$20,000 in 2019 dollars, below which reported money income does not reliably reflect resources, though those households are retained in

the population counts.

I apply five restrictions, which together reduce the sample to 1,595 block groups.

1. *Student and retirement communities.* I drop block groups in which more than 75 percent of residents are aged 18–24 or more than 75 percent are aged 65 and over. In these places residence is tied to a university or an age-restricted development rather than to the neighborhood choice the model represents, and Census money income omits the transfers, scholarships, and retirement drawdowns on which these residents rely.
2. *Vacation-home enclaves.* I drop block groups in which seasonally vacant units exceed 90 percent of the vacant stock (B25004), a signature of second-home concentration in which recorded resident income does not reflect the population that occupies the housing.
3. *Exurban fringe.* I drop block groups in the top one percent by land area or the top one percent by distance from the downtown Phoenix core, together with the bottom one percent by population. Because the Census draws block groups to roughly equal population, extreme land area signals very low density at the urban edge, whose residents do not sort on the margins the model represents.
4. *Ultra-luxury enclaves.* I drop the top decile of block groups by minimum-lot cost R_n . In these enclaves the true income of high-group residents exceeds the capped top income bracket, so recorded income understates the resources of the households actually sorting there and would distort the calibration.
5. *Degenerate block groups.* I drop block groups in which fewer than two income groups are present or fewer than 25 households reside, since the sorting problem is not defined where an income group is absent and shares are unreliable where the population is very sparse.

C Unregulated-Price Calibration

In block groups with both regulated and unregulated zones, I hold the estimated regulated price P_n^{reg} fixed and recover the unregulated price P_n^{unreg} from the requirement that the model’s regulated population share equal the observed regulated unit share s_n^{reg} :

$$\sum_w \omega_{wn} s_{\text{reg}|wn}(P_n^{\text{reg}}, P_n^{\text{unreg}}) = s_n^{\text{reg}}, \quad (23)$$

where $s_{\text{reg}|wn}$ is the within-neighborhood probability that a group- w household chooses the regulated zone, given by Equation (9), and ω_{wn} is the population weight of group w in block group n . Raising the unregulated price makes the regulated zone relatively more attractive and shifts population into it. I solve Equation (23) for P_n^{unreg} by bisection. The minimum lot lowers the value of the regulated option by forcing its residents above their preferred housing consumption, so P_n^{unreg} clears above P_n^{reg} by an amount increasing in the regulatory distortion, and equals it when no minimum lot binds. Where a block group contains only one zone the clearing condition is degenerate, and I set the single price equal to the estimated index, $P_n^{\text{unreg}} = P_n^{\text{reg}}$.

D The Gradient Alignment Instrument

The instrument Γ_n exploits nighttime heat advection: near-surface airflow transports hot or cool air from one micro-climate into adjacent block groups, creating temperature differences that are not mechanically tied to a block group’s own landscaping or built form. I construct it in two steps.

First, I simulate near-surface (5 m) backward air-parcel trajectories for July nighttime hours (6 pm, 12 am, 6 am) over Maricopa County using NOAA’s HYSPLIT model driven

by the High-Resolution Rapid Refresh (HRRR) atmospheric dataset. For each block group these trajectories identify the predominant bearings along which nighttime air arrives.

Second, around each block-group centroid I partition the surrounding ring into eight directional slices (N, NE, . . . , NW) and compute, for each slice, its July nighttime temperature anomaly α_{nd} relative to the ring mean, from the HUMID temperature surface. I then measure how closely the prevailing airflow aligns with hotter versus cooler slices:

$$\Gamma_n = \sum_d \omega_{nd} \left(1 - \frac{\min\{|\vartheta_d - \vartheta_n|, 360^\circ - |\vartheta_d - \vartheta_n|\}}{180^\circ} \right) \alpha_{nd}, \quad (24)$$

where ϑ_n is the dominant trajectory bearing, ϑ_d the bearing of slice d , ω_{nd} the frequency weight on that bearing, and α_{nd} the normalized temperature anomaly of slice d . A positive Γ_n means the block group sits downwind of hotter-than-average land; a negative value means it sits downwind of cooler-than-average land. Because Γ_n depends on airflow and on the temperature of land *outside* the block group’s own boundary, but not on its own built environment, it shifts Temp_n in a way plausibly orthogonal to unobserved neighborhood quality. I match the ring radius to tract density, using a 500 m ring in the denser half of tracts and a 1000 m ring in the sparser half, so the external patches enter at the scale at which they influence a block group’s temperature.

E Minimum-Lot Crosswalk

I measure the minimum lot ℓ_n primarily from authoritative zoning. For Phoenix, Mesa, Chandler, Gilbert, Glendale, Queen Creek, Scottsdale, Tempe, and unincorporated Maricopa County, I obtain each jurisdiction’s residential zoning GIS layer and map every zoning code to the minimum lot size published by that jurisdiction, in square feet. Multifamily, townhome, two-family, and manufactured-housing districts are assigned a minimum of zero

and constitute the model’s unregulated zone. I overlay the combined zoning layer on the block-group geography and assign each block group the minimum lot of the zoning district covering the largest share of its area.

For less-populated municipalities without a usable coded layer, I estimate minimum lot sizes from the modal single-family parcel size in the block group, estimated by a kernel-density mode of parcel acreage from Assessor sales. I then reconcile the two: where an authoritative zoning minimum exceeds twice the observed sold-lot mode and the block group has at least thirty single-family sales, I use the observed mode, since in fringe blocks a large-lot rural or environmentally-constrained polygon of mostly undeveloped desert can otherwise be stamped onto a block whose actual stock sits on much smaller lots. Block groups with neither an authoritative minimum nor a reliable transaction mode are imputed at the county median. Zones assigned a minimum of zero are never imputed, so that unregulated block groups are preserved.

F Cooling Delivered by the Cool Corridors

I construct the cooling delivered by the Cool Corridors from the program’s own planting design rather than from an assumed canopy target, so that each step is tied to a documented parameter.

Trees planted. The program plants at 200 trees per mile of corridor (City of Phoenix, 2024a). Because the corridors are recorded as right-of-way polygons rather than centerlines, I recover each corridor’s length from its area A and perimeter P by solving for the rectangle with the same area and perimeter, $L = (P + \sqrt{P^2 - 16A})/4$ and $W = (P - \sqrt{P^2 - 16A})/4$. The recovered widths have a median of roughly 41 metres, consistent with a Phoenix arterial right-of-way, which supports the elongated-polygon approximation. Applying the planting

density to the recovered length gives each corridor’s tree count; I assume all planted trees become established.

From trees to canopy area. The five most frequently planted species are Mulga, Ghost Gum, Southern Live Oak, Chinese Elm, and Blue Palo Verde (City of Phoenix, 2024a), corresponding by mature stature to the small, large, medium, large, and medium size classes of McPherson et al. (2004), whose growth curves report established crown spreads of 22, 36, and 41 feet for small, medium, and large street trees, implying crown areas of 35.3, 94.6, and 122.7 square metres. These require no adjustment for street conditions, since the curves are estimated from street and park trees in Glendale, Arizona. I average crown *areas* rather than diameters, because area is convex in diameter; weighting the five species equally gives a mean established crown area of 94.0 square metres and a mean crown radius of 5.47 metres. A corridor’s added canopy area is its tree count times this mean.

From canopy area to canopy cover. I buffer each corridor polygon by the mean crown radius before intersecting it with the block-group geography, so that canopy overhanging the right-of-way is assigned to the block group it shades; this matters because corridors follow arterials, which often form block-group boundaries. Canopy is allocated to block groups in proportion to each corridor’s buffered footprint within them. The assumed corridor width cancels from the cooling calculation through an identity, the block group’s corridor share times the on-corridor canopy share equals the block-group canopy increase, and affects only which block groups are treated.

From canopy cover to temperature. I map the block-group canopy increase into an air-temperature reduction using the ENVI-met simulations of Middel et al. (2014) for a typical Phoenix residential neighborhood, which report reductions of 3.6 °F as canopy rises from zero to 10 percent and a further 4.3 °F from 10 to 25 percent. Treating this as piecewise-linear

implies marginal cooling of 36.0°F per unit of canopy over the first segment and 28.7°F over the second. I evaluate the cooling delivered to each block group as the dose-response at its post-planting canopy minus its pre-planting canopy, the latter from documented low corridor-neighborhood canopy shares (City of Phoenix, 2024c). Because the underlying estimates are neighborhood-scale, applying them to a block-group canopy change avoids the scale mismatch of applying a neighborhood-scale response to a narrow strip. The resulting reduction enters the amenity function through the income-group-specific marginal value of cooling θ_w .

Sensitivity. The cooling is linear in crown area and hence quadratic in crown spread, the most consequential parameter. I report robustness to the 22–41 foot range of the size classes and to the assumption of full survival, against which McPherson et al. (2004) assumes 40 percent of planted trees are lost over forty years.

Binding regulated zone

When the minimum lot binds, utility is $V_{\text{reg},wn}^{\text{bind}} = \log a_{wn} + B_{wn}D_{wn}$, with the undistorted consumption value and distortion factor

$$B_{wn} \equiv k_w \frac{I_w - P_n^{\text{reg}} \bar{h}}{[P_n^{\text{reg}}]^\beta}, \quad D_{wn} \equiv \kappa \frac{(R_n - P_n^{\text{reg}} \bar{h})^\beta (I_w - R_n)^{1-\beta}}{I_w - P_n^{\text{reg}} \bar{h}}, \quad D_{wn} \leq 1.$$

Cooling raises the per-unit price P_n^{reg} and, through it, the mandated lot cost R_n . Differentiating with respect to one degree of cooling gives three channels,

$$\Delta V_{\text{reg},wn}^{\text{cool}} = \underbrace{\theta_w}_{\text{direct benefit}} - \underbrace{D_{wn} \Phi_{wn}^{\text{reg}}}_{\text{rent penalty, moderated by the distortion}} + \underbrace{B_{wn} \frac{dD_{wn}}{d \log a_{wn}}}_{\text{change in the distortion}}, \quad (25)$$

where Φ_{wn}^{reg} is the rent penalty of (22) evaluated at the regulated price. The second term is smaller than the unregulated rent penalty because $D_{wn} \leq 1$: a household already forced above its optimal housing expenditure absorbs less additional loss from a price increase, since a portion of that expenditure was already mandated by the lot. The third term is the change in the distortion itself,

$$\frac{dD_{wn}}{d \log a_{wn}} = D_{wn} \left[\beta \frac{R'_n(L_n) - P_n^{\text{reg}'}(L_n) \bar{h}}{R_n - P_n^{\text{reg}} \bar{h}} - (1 - \beta) \frac{R'_n(L_n)}{I_w - R_n} + \frac{P_n^{\text{reg}'}(L_n) \bar{h}}{I_w - P_n^{\text{reg}} \bar{h}} \right],$$

where $R'_n(L_n)$ and $P_n^{\text{reg}'}(L_n)$ denote the response of the lot cost and the regulated price to the change in land value L_n induced by cooling. Its sign is ambiguous in general, but

$$\lim_{I_w \rightarrow R_n^+} \frac{dD_{wn}}{d \log a_{wn}} < 0.$$

For a household whose income is close to the minimum lot cost, the amenity increase makes the lot more distortionary, because the household retains little income after meeting the mandated housing expenditure, so the rent increase pushes the ratio of constrained to unconstrained utility further below one. The third channel is therefore an additional loss for exactly the low-income households on whom the lot binds.

Implication

Taken together, (22) and (25) imply that a cooling improvement reduces welfare whenever the rent penalty exceeds the direct benefit θ_w , and that the penalty is larger for low-income households in both zones, because it scales with the housing expenditure share. In a binding regulated zone the third channel adds to the loss for households whose income is near R_n . A microclimate improvement is thus most likely to harm low-income households in regulated neighborhoods, which is the force behind the regressive incidence of cooling reported in Section 6.2.1 and the motivation for pairing cooling with zoning reform.

G Homothetic demand and the incidence of amenity capitalization

This appendix derives the two results stated in Sections 3.6.1 and 3.6.2. Throughout, I_w is the income of group w , P_n^o the price of housing in zone o of neighborhood n , $\bar{h}$ the subsistence quantity of housing, R_n the dollar cost of a minimum lot in the regulated zone of n , and β the Stone-Geary weight. Cobb-Douglas is the special case $\bar{h} = 0$.

G.1 The incidence of a rent increase

Omitting neighborhood and zone subscripts, as they are not needed, the household solves

$$\max_{h,g} (h - \bar{h})^\beta g^{1-\beta} \quad \text{subject to} \quad Ph + g \leq I_w,$$

and, in the unregulated zone or where the minimum lot is slack, the solution is

$$h_{w,n}^* = \bar{h} + \frac{\beta}{P}(I_w - P\bar{h}), \quad g_{w,n}^* = (1 - \beta)(I_w - P\bar{h}),$$

with indirect utility

$$V(P, I_w) = \Lambda P^{-\beta} (I_w - P\bar{h}), \quad \Lambda \equiv \beta^\beta (1 - \beta)^{1-\beta}.$$

Proposition 1. *The money-metric welfare loss from a marginal increase in the price of housing, expressed as a share of income, equals the housing expenditure share:*

$$-\frac{1}{I_w} \cdot \frac{\partial V / \partial \log P}{\partial V / \partial I_w} = \frac{Ph_{w,n}^*}{I_w} = \beta + (1 - \beta) \frac{P\bar{h}}{I_w}.$$

This is strictly decreasing in income whenever $\bar{h} > 0$ and is independent of income when

$$\bar{h} = 0.$$

Proof. Differentiating, $\partial V/\partial I_w = \Lambda P^{-\beta}$, which does not depend on income. Differentiating with respect to price and collecting terms,

$$\frac{\partial V}{\partial P} = -\beta \Lambda P^{-\beta-1} (I_w - P\bar{h}) - \Lambda P^{-\beta} \bar{h} = -\Lambda P^{-\beta} \left[\bar{h} + \frac{\beta}{P} (I_w - P\bar{h}) \right] = -\frac{\partial V}{\partial I_w} h_{w,n}^*,$$

which is Roy's identity. which is Roy's identity. Since $\partial V/\partial \log P = P \partial V/\partial P = -(\partial V/\partial I_w) P h_{w,n}^*$, dividing through by $\partial V/\partial I_w$ and then by I_w gives the housing expenditure share $P h_{w,n}^*/I_w$. Differentiating that share with respect to income yields $-(1 - \beta)P\bar{h}/I_w^2$, which is strictly negative for $\bar{h} > 0$ and zero otherwise. $\square$

The compensating payment required to restore a household's utility after a price increase is the number of units of housing it consumes, multiplied by the price change. Re-optimization is not included in the effect because by the envelope theorem, the household was already at an optimum and the gain from adjusting is second order in dP . The incidence of a rent increase therefore depends only on how much housing the household holds as a share of income, and under Stone-Geary that quantity rises less than proportionally with income.

G.2 Minimum lots as a regulatory subsistence requirement

Where the minimum lot binds, the constraint $P_n^{reg}h \geq R_n$ holds with equality and the household consumes $h = R_n/P_n^{reg}$ and $g = I_w - R_n$. It is no longer on its demand curve. Indirect utility becomes

$$V^{\text{bind}}(P, I_w) = \left(\frac{R_n}{P} - \bar{h} \right)^\beta (I_w - R_n)^{1-\beta}.$$

Proposition 2. *Roy's identity continues to hold at the constrained optimum, and as a share*

of income the money-metric loss from a marginal price increase is

$$-\frac{1}{I_w} \cdot \frac{\partial V^{\text{bind}}/\partial P}{\partial V^{\text{bind}}/\partial I_w} = \frac{R_n}{I_w},$$

which is strictly decreasing in income for any β , including under homothetic preferences.

Proof. Write $x \equiv R_n/P - \bar{h}$ and $y \equiv I_w - R_n$, so that $V^{\text{bind}} = x^\beta y^{1-\beta}$. Then $\partial V^{\text{bind}}/\partial I_w = (1 - \beta)x^\beta y^{-\beta}$ and

$$\frac{\partial V^{\text{bind}}}{\partial P} = \beta x^{\beta-1} y^{1-\beta} \cdot \left(-\frac{R_n}{P^2} \right) = -\frac{\beta}{1-\beta} \cdot \frac{R_n}{P^2} \cdot \frac{y}{x} \cdot \frac{\partial V^{\text{bind}}}{\partial I_w}.$$

Substituting $x = R_n/P - \bar{h}$ and simplifying, the ratio reduces to the quantity of housing consumed, R_n/P , so that $\partial V^{\text{bind}}/\partial P = -(R_n/P) \partial V^{\text{bind}}/\partial I_w$, which is Roy's identity with $h = R_n/P$. Multiplying by P and dividing by I_w gives R_n/I_w . Since R_n is a property of the neighborhood and does not vary with w , the share is strictly decreasing in I_w . $\square$

Propositions 1 and 2 together establish that $\bar{h}$ and R_n act similarly as a housing consumption floor. Each compels the household to hold a fixed quantity of housing that does not scale with income, and each therefore generates a housing burden that falls with income. The first is a property of preferences and the second a property of the zoning code. In this sense land-use regulation is a substitute for non-homothetic demand in producing a regressive incidence of housing costs. Under Cobb-Douglas the regulation supplies the entire gradient. Removing the minimum lot would collapse the Cobb-Douglas burden gradient to unity while leaving the Stone-Geary gradient at roughly three.

However, this effect is only partial because the constraint binds on a subset of households. The regressive share R_n/I_w obtains only where the lot binds and where it is slack, the household consumes its optimum housing demand and the incidence reverts to the expenditure share b_{wn} of Proposition 1. Since the minimum lot binds on about a third of low-income

Cool Corridor renters and fewer at higher incomes the homothetic model displays slight regressivity only at the lowest income group, where binding is most common, rather than the continuous gradient non-homothetic demand produces everywhere. My results express this, as under Cobb-Douglas the welfare cost of the cooling-induced rent increase is almost flat, 0.83 percent of income for the lowest group and about 0.67 percent from the lower-middle group upward, whereas under Stone-Geary it falls steadily from 1.18 percent at the bottom to 0.45 percent at the top. The regulation is thus the homothetic model's only channel for regressive incidence, but because it reaches only the constrained minority it reproduces a very small regressive incidence, instead of the significantly larger distribution-wide gradient that non-homothetic preferences generate.

G.3 The estimated value of cooling

The amenity is recovered by inverting observed location choices. The probability that a type- w household lives in neighborhood n is

$$\pi_{wn} = \frac{\exp(\rho [\nu_{wn} + \log a_{wn}])}{\sum_m \exp(\rho [\nu_{wm} + \log a_{wm}])},$$

where ν_{wn} is the money-metric consumption value per dollar of income of the bundle attainable in neighborhood n , and ρ is the across-neighborhood migration elasticity. Inverting,

$$\log a_{wn} = \frac{1}{\rho} \log \pi_{wn} - \nu_{wn} + \text{constant}(w). \quad (26)$$

Because the amenity is a residual, its value moves inversely with the consumption value. A neighborhood where a group gets little consumption value for its income must carry a high amenity to explain why the group lives there. A neighborhood where the consumption value is high relative to income can carry a low one, because consumption already accounts for the choice.

Assumption 1. Cooler neighborhoods are more expensive. A neighborhood one degree warmer costs a fraction c less, so that $\partial \log P_n^o / \partial T_n = -c$ with $c > 0$.

Proposition 3. *Under Assumption 1, the estimated marginal willingness to pay for cooling of group w satisfies*

$$\theta_w \equiv -\frac{\partial \log a_{wn}}{\partial T_n} = \zeta_w + c b_{wn}, \quad \zeta_w \equiv -\frac{1}{\rho} \frac{\partial \log \pi_{wn}}{\partial T_n},$$

and consequently

$$\left[\theta_H^{CD} - \theta_L^{CD} \right] - \left[\theta_H^{SG} - \theta_L^{SG} \right] = c [b_L - b_H] > 0. \quad (27)$$

Homothetic demand overstates the spread in the estimated value of cooling between low- and high-income households, by the price-temperature gradient times the gap in housing expenditure shares. The bias disappears only when the true demand is homothetic.

Proof. Differentiate the inversion (26) with respect to temperature. Since temperature only affects ν_{wn} through price, and $\partial \nu_{wn} / \partial \log P_n^o = -b_{wn}$ by Proposition 1,

$$\begin{aligned} \theta_w &\equiv -\frac{\partial \log a_{wn}}{\partial T_n} = -\frac{1}{\rho} \frac{\partial \log \pi_{wn}}{\partial T_n} + \frac{\partial \nu_{wn}}{\partial T_n} \quad (\text{differentiate (26), flip sign}) \\ &= \zeta_w + \frac{\partial \nu_{wn}}{\partial \log P_n^o} \cdot \frac{\partial \log P_n^o}{\partial T_n} \quad (\text{definition of } \zeta_w; \text{ chain rule through price}) \\ &= \zeta_w + (-b_{wn})(-c) \quad (\text{Proposition 1; Assumption 1}) \\ &= \zeta_w + c b_{wn}. \end{aligned}$$

The first term collects everything temperature does through location choices; the second is the budget channel, in which a warmer neighborhood is cheaper and the price relief is worth more to a household with a larger housing share, the two signs leaving a positive $c b_{wn}$. ζ_w depends only on the observed distribution of population across neighborhoods and

temperature, and since both demand systems are fit to the same populations, it is common to both and cancels when they are compared across income groups, so the two differ only through b_{wn} . Differencing across income groups,

$$\theta_H - \theta_L = [\zeta_H - \zeta_L] + c[b_H - b_L].$$

Under Cobb-Douglas $b_H = b_L = \beta$ for unconstrained households, so the second term is zero. Under Stone-Geary $b_H < b_L$ by Proposition 1, so it is negative. Subtracting the two gives (27), positive because $c > 0$ and $b_L > b_H$. $\square$

Low-income households are concentrated in hot, inexpensive neighborhoods. Stone-Geary can attribute much of that concentration to the budget, because a household spending forty three percent of its income on housing is strongly deterred by the price of the cooler alternative. Cobb-Douglas cannot, because under homothetic preferences the proportional cost of a price increase is the same at every income, so it instead concludes that low income households stay in hot neighborhoods because they have a lower aversion to hot temperatures. This bias runs in the opposite direction at the top of the income distribution. Cobb-Douglas assigns high-income households a larger price deterrent than they actually face, and so reads their presence in expensive, cooler neighborhoods as a stronger distaste for heat than the non-homothetic model finds. Low-income preferences for cooling are understated, and high-income preferences are overstated relative to non-homothetic demand.

G.4 Decomposing the regressivity of urban cooling

Consider a household in a cooled neighborhood. The money-metric value of the amenity improvement is $I_w \theta_w \Delta T$ and the rent that follows it costs $Ph_{w,n} \Delta \log P = b_{wn} I_w \Delta \log P$.

The share of the amenity value that survives the price response is therefore

$$\mathcal{R}_w = 1 - \kappa \frac{b_{wn}}{\theta_w}, \quad \kappa \equiv \frac{\Delta \log P}{\Delta T},$$

with κ common to all households in a neighborhood. Since income does not appear, retention only depends on the ratio between the housing expenditure share to the and the marginal willingness to pay for cooling, allowing the regressivity of an amenity improvement to reflect two effects:

$$\underbrace{\frac{b_L/\theta_L}{b_H/\theta_H}}_{\text{regressivity}} = \underbrace{\frac{b_L}{b_H}}_{\text{budget channel}} \times \underbrace{\frac{\theta_H}{\theta_L}}_{\text{preference channel}}.$$

	Budget channel, b_L/b_H		Preference channel
	Observed	Absent the minimum lot	θ_H/θ_L
Stone-Geary	$0.428/0.137 = 3.12$	$0.429/0.136 = 3.17$	$0.138/0.082 = 1.68$
Cobb-Douglas	$0.329/0.248 = 1.33$	$\beta/\beta = 1.00$	$0.151/0.075 = 2.01$

Total regressivity is 5.26 under Stone-Geary and 2.67 under Cobb-Douglas, so the homothetic specification recovers roughly half of it. However the half it does recover is attributed to the wrong sources. The preference channel estimated by the Cobb-Douglas model, 2.01, exceeds the true value of 1.68 implied by Stone-Geary (by Proposition 3, the homothetic model attributes poor household's concentration in hot neighborhoods to a weaker preference for cooling rather than as an affordability issue). The small budget channel produced by Cobb-Douglas, 1.33, is manufactured entirely by the minimum lot, and without the constraint it would collapse to unity. Under Stone-Geary demand, the budget channel is much larger (3.12). The two specifications therefore diverge in conclusions regarding what benefits accrue to renters under zoning reform. The homothetic model concludes that removing the constraint equalizes the incidence of cooling across income, save the preference gap, when

the affordability gradient is a property of demand and reform strips out only the, small regulation-generated component the homothetic model reproduces.

G.5 Populations by Income Group

Table 7 reports the number of households in each income group, both countywide and within the 179 Cool Corridor block groups, split by tenure. Two features of the composition impact results. The corridors are poorer than the county as a whole: low-income households are 29 percent of corridor residents against 23 percent countywide, while high-income households are 23 percent of the corridor against 31 percent countywide. Tenure varies substantially with income inside the corridors, from 64 percent renters among low-income households to 81 percent owners among high-income households, which is why the incidence of a cooling-induced rent increase depends so heavily on the renter share of each group.

Table 7: Household counts by income group, countywide and in the Cool Corridors

Income group	Countywide			Cool Corridors		
	Renters	Owners	Total	Renters	Owners	Total
Low (< 44,100)	131,541	99,105	230,646	19,672	11,226	30,898
Lower-middle (44,100–75,600)	93,481	113,855	207,336	12,891	11,954	24,845
Upper-middle (75,600–126,000)	80,853	170,145	250,998	9,313	15,855	25,168
High (> 126,000)	48,162	261,744	309,906	4,692	19,508	24,200
Total	354,037	644,849	998,886	46,568	58,543	105,111

Notes. Baseline household counts in the calibrated model, by income group and tenure. “Countywide” includes all 1,595 block groups in the estimation sample; “Cool Corridors” covers the 179 block groups overlapping a Cool Corridor. Income ranges are annual household income in 2025 dollars. Counts are household populations and are held fixed across counterfactuals. Tenure is assigned after households sort, as described in Section 3.5.

H Aggregate policy accounting

The welfare results in the main text are per household. In this appendix I aggregate them. Table 8 multiplies each group's per-household equivalent variation, and each of its three components, by the number of renter households in that group, so the entries are annual flows in dollars and they add across groups. The county has roughly 354,000 renter households, of whom 131,500 are in the lowest income group and 48,200 in the highest.

Cooling alone creates \$40.7 million a year of amenity value and delivers \$34.9 million of it, the difference of \$5.8 million being the additional rent that renters pay once prices adjust. Pairing the canopy with zoning reform in forty block groups raises the net annual gain to \$59.6 million, an increase of \$24.7 million, and the accounting shows that the increase arrives through two distinct channels. The first is \$11.1 million of value created by relaxing the minimum lot, which is surplus that did not previously exist and is not taken from anyone, and \$8.3 million of it accrues to the lowest income group against nothing at all to the highest. The second is the reversal of the rent transfer. Under the canopy alone, renters send \$5.8 million a year to landowners; under the canopy paired with zoning reform, they receive \$7.8 million, because the relaxed minimum lot lowers the price of housing services. The swing is \$13.6 million a year, and since rental housing in the model is held by absentee owners, it is money that would otherwise have left the county.

The aggregate figures remain regressive. The Cooling policy alone delivers \$16.5 million to 48,200 high-income renter households and \$5.1 million to 131,500 low-income households. The high income group receives three times the gains and comprises a third as many households as the low income group.

Table 8: Citywide policy accounting, renter households (\$ millions per year)

	Low	Lower-middle	Upper-middle	High	Total
Cooling alone					
Value of the tree canopy	7.8	6.7	9.6	16.7	40.7
Rent transferred to landowners	-2.6	-1.6	-1.3	-0.2	-5.8
Net gain to renters	5.1	5.0	8.3	16.5	34.9
Cooling and zoning reform in forty block groups					
Value of the tree canopy	7.8	6.7	9.6	16.7	40.7
Value created by relaxing the minimum lot	8.3	2.4	0.4	0.0	11.1
Rent transferred to landowners	2.0	2.0	1.9	2.0	7.8
Net gain to renters	18.1	11.0	11.9	18.7	59.6
<i>Memo:</i> renter households (thousands)	131.5	93.5	80.9	48.2	354.0

Notes: All renter households in Maricopa County, across all 1,595 block groups, measured against the HB 2721 policy baseline. Figures are annual flows in millions of dollars, aggregated from the per-household equivalent variation of Appendix 3 and its three components, summing across income groups. Table 5.

I The incidence of the minimum lot

While the minimum lot rule applies to every household in a regulated block group, its incidence is not the same across income groups. Table 9 reports the share of households for whom the required lot exceeds the housing they would otherwise choose, at the calibrated equilibrium. Countywide, the constraint binds on 46.9 percent of low-income households and on 0.33 percent of high-income households. High income households are almost never constrained by a minimum lot, because it already demands more housing than any minimum lot the sample requires.

Inside the Cool Corridors minimum lots bind for 33.4 percent of low-income households and for zero high-income households, because corridor block groups are denser and more central than the county as a whole and their minimum lots are correspondingly smaller, though still restrictive to the poorest households. Relaxing the minimum lot can only return value to households who faced binding constraints at the benchmark, so the value of removing the zoning distortion reported in Table 3 is \$444 a year for a low-income household and zero for a high-income households.

Table 9: Percent population facing binding minimum lot constraints

Income group	Income (\$)	Households on whom the minimum lot binds (%)	
		Countywide	Cool Corridors
Low	34,793	46.91	33.40
Lower-middle	59,529	38.07	21.26
Upper-middle	99,388	23.39	9.13
High	209,273	0.33	0.00

Notes: Share of households in regulated housing for whom the minimum lot requirement exceeds the housing they would otherwise choose, at the calibrated equilibrium, weighted by population. A household is counted as constrained when the required lot exceeds unconstrained Stone-Geary demand.

J Cooling and zoning interactions

Table 10 enacts each policy alone and then both together. The interaction, defined as the equivalent variation of the two together less the equivalent variation of each alone, is positive for every income group, so the two policies are complements rather than substitutes. Enacting cooling along with zoning reform jointly is worth more than the sum of enacting each on its own. Decomposing the interaction into the same three channels as Table 3 helps define the mechanism of the complementarity. Most of the effect originates from the price effect, at \$12.08 of the \$9.22 a low-income household gains. The price damage of enacting both policies together is smaller than the sum of the price damage of each policy alone. The housing freed up when constrained households are permitted to buy less absorbs part of the demand the cooling amenity creates, so new supply meets new demand in the same block groups at the same time.

The complementarity is concentrated at the bottom of the distribution, at \$9.22 for a low-income household against \$0.40 for a high-income one, because the targeted block groups were selected where the minimum lot was costing low-income households the most. The price relief therefore accrues to households the cooling amenity was raising rents on.

Table 10: Measuring policy complementarity

	Low	Lower-middle	Upper-middle	High
<i>Equivalent variation (\$/year)</i>				
Cooling alone	20.17	107.35	522.51	3,330.87
Zoning reform alone	477.06	258.29	115.14	63.35
Both together	506.46	377.93	645.17	3,394.63
Interaction	9.22	12.29	7.52	0.40
<i>Source of the interaction (\$/year)</i>				
Cooling amenity	0.00	0.00	0.00	0.00
Value of removing the zoning distortion	-2.87	-3.07	-0.30	0.00
Net price effect	12.08	15.35	7.82	0.40

Notes: Annual welfare flows for renter households in the 179 block groups that overlap a Cool Corridor, measured against the HB 2721 policy baseline. The interaction is the equivalent variation of the two policies enacted together less the equivalent variation of each enacted alone. A positive value implies the policies are complements. It is decomposed into the same three channels as Table 3.

J.1 Binding Minimum-Lot Constraints by Demand System

The two demand systems produce large differences in the population share by income group facing binding minimum lot requirements in the cool corridor neighborhoods, with discrepancy concentrated at the bottom of the income distribution. Table 11 reports, for each income group, the share of Cool Corridor households in regulated housing for whom the required minimum lot exceeds the housing they would otherwise choose. Under non-homothetic Stone-Geary demand the lot binds on about a third of low-income corridor households; under homothetic Cobb-Douglas demand it binds on nearly three-quarters. The gap arises because Cobb-Douglas assigns every household the same desired housing share, $\beta = 0.248$, well below the share a low-income household actually chooses, so many lots that a low-income household would exceed on its own are counted as binding. The ordering reverses in the upper-middle group, where Stone-Geary implies a desired share below β and therefore finds the lot binding slightly more often. Because the value of relaxing the minimum lot accrues only to households the lot actually constrains, this difference is the source of the homothetic model’s overstatement of the gains from zoning reform (Section 6.2.4).

Table 11: Share of Cool Corridor households facing a binding minimum lot, by income group and demand system

Income group	Households facing a binding lot (%)	
	Stone-Geary	Cobb-Douglas
Low (< 44,100)	34.5	71.9
Lower-middle (44,100–75,600)	22.6	31.3
Upper-middle (75,600–126,000)	9.4	6.7
High (> 126,000)	0.0	0.0

Notes. Population-weighted share of households in regulated housing within the 179 Cool Corridor block groups for whom the required minimum lot exceeds the housing they would choose absent the constraint, by income group, under each demand system. Both models are calibrated to the same populations and face the same prices; they differ only in the desired housing share, which Stone-Geary lets vary with income and Cobb-Douglas fixes at $\beta = 0.248$. Rates are evaluated at the cooling counterfactual; the Stone-Geary figures run about one percentage point above the pre-cooling baseline rates reported in Table 9.

J.2 Zoning Reform under Non-homothetic and Homothetic Demand

Table 12 repeats the welfare decomposition in the main text for the combined policy, cooling paired with minimum-lot relaxation in the forty most-constrained corridor block groups, under each demand system. The homothetic model heavily overstates gains to renters from zoning reform compared to Stone-Geary demand (\$1,156 against \$444). The same pattern appears in the rent burden, where the homothetic model records the reform lowering the low-income housing cost burden by 3.4 percentage points, against 0.6 under the non-homothetic model, because it treats low-income households as forced far above their desired housing and so credits zoning reform with much higher efficiency gains. Table 13 links this distortion to differences in the extent of zoning reform that a planner would choose.

Table 12: Cooling paired with targeted zoning reform under non-homothetic and homothetic demand, across the income distribution

Household income (2025 \$)		Low < 44,100	Lower-middle 44,100–75,600	Upper-middle 75,600–126,000	High > 126,000
Panel A. Welfare of the combined policy, \$/year (% of income)					
Value of cooling	SG	432 (1.24)	584 (0.98)	1,155 (1.16)	4,281 (2.05)
	CD	395 (1.14)	584 (0.98)	1,251 (1.26)	4,692 (2.24)
Value of removing the zoning distortion	SG	444 (1.28)	203 (0.34)	52 (0.05)	0 (0.00)
	CD	1,156 (3.32)	261 (0.44)	4 (0.00)	0 (0.00)
Net price effect	SG	−369 (−1.06)	−409 (−0.69)	−562 (−0.57)	−886 (−0.42)
	CD	−197 (−0.57)	−316 (−0.53)	−568 (−0.57)	−1,224 (−0.58)
Total welfare (EV)	SG	506 (1.46)	378 (0.63)	645 (0.65)	3,395 (1.62)
	CD	1,353 (3.89)	530 (0.89)	687 (0.69)	3,468 (1.66)
Panel B. Housing cost burden (rent as % of income)					
Baseline	SG	39.4	26.5	19.2	14.0
	CD	30.6	26.4	24.9	24.8
With combined policy	SG	38.8	26.2	19.3	14.2
	CD	27.2	25.2	24.8	24.8
Change (p.p.)	SG	−0.6	−0.3	+0.1	+0.2
	CD	−3.4	−1.2	−0.1	+0.0

Notes. Renter households in the 179 Cool Corridor block groups, measured against the HB 2721 baseline, under cooling combined with minimum-lot relaxation in the forty highest-ranked block groups. Both models face the identical policy and are calibrated to the same populations; they differ only in the demand system. Panel A reports each household’s equivalent variation and its three components, which sum to the total, with the same quantity as a share of group income in parentheses. Panel B reports the housing cost burden, rent paid as a share of income. Sorting under the combined policy is reported for the non-homothetic model in Table 3. Each model selects its own forty block groups on the low-income improvement measure of Section 5.1.

Table 13: Returns to low income renters as zoning reform is scaled: non-homothetic versus homothetic housing demand

Block groups upzoned	Stone-Geary		Cobb-Douglas	
	Benefit (\$)	% of max	Benefit (\$)	% of max
10	276	55	627	39
20	408	81	918	57
40	486	96	1,248	78
60	503	99	1,443	90
80	506	100	1,555	97
100	506	100	1,607	100

Notes: Benefit of the zoning reform to a low-income renter household in the Cool Corridors, measured as the increment to annual equivalent variation from adding the reform to cooling, against the HB 2721 baseline. “% of max” expresses each row relative to the value at one hundred upzoned block groups.